

IDENTIFICATION

Product Code:	MAINDEC-8I-D6AB-D
Product Name:	AX08 Diagnostic
Date Created:	October 8, 1968
Maintainer:	Diagnostic Group
Author(s):	J. Kelly J. Hittell

1. ABSTRACT

This unit is tested in three sections: (a) an instruction test of the logic; (b) a display test for the scope; (c) a calibration section for the A/D Converter.

2. REQUIREMENTS

2.1 Equipment

PDP-8, 8L or 8I Standard Computer

AX08 option

Adjustable Voltage Source (0.01% or better, $Z_{out} < 1.0 \text{ ohm}$)

2.2 Storage

2.2.1 Program Storage - The routine uses memory from address 0 to 4500.

3. LOADING PROCEDURE

3.1 Methods

Procedure for normal binary tape is followed.

4. STARTING PROCEDURE

For normal starting operation all switches should be down. Starting address is 200.

4.1 Program and/or Operator Action

Connect non-zero voltage source to input connector for channel zero.

Set "Timing Control" minimum (C.C.W.)

Load the program into memory.

Set switch register to starting address - SA = 0200

Load Address.

Press Start.

4.1.1 The program will loop in the first section of the test, and the display, if on, will read
"DIAGNOSTIC RUNNING"
"SAME VALUE IN XX ICMX"

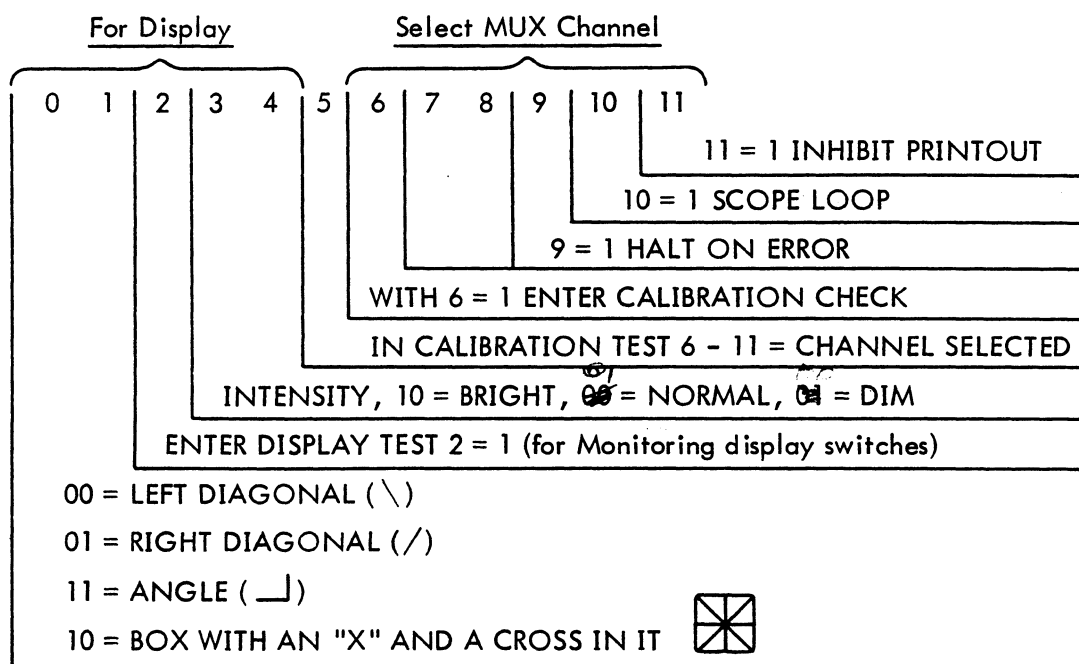
If there is an error, the teleprinter will print out the error and continue on in section test. (XX = the multiplexer channel + 1 where overflow occurs)

5. OPERATING PROCEDURE

5.1 Operational Switch Settings

With all switches down (Logical zero), the test will stay in the instruction section. With switch 2 up the test will enter the display section, from the instruction section. With switch 6 up the test will enter the A/D calibration section, from the display section.

5.1.1 Switches -



5.2 Subroutine Abstract

5.2.1 BEGIN - This is AX08 instruction test of the logic which tests that all registers and flip-flops initiated by start key are in the correct state. Also tested is the ability to set and clear the (Y) register, the skip on flag and interrupt logic for--the A/D timing, the RC clock and the crystal clock. Other logic functions of the display and the A/D converter are also tested.

5.2.2 CHTST - This is an increment test of multiplexer and assumes that (1) the display is operating; (2) the A/D is calibrated; (3) the channel zero of the multiplexer has a non-zero voltage applied to it,

and that all other channels have no voltage applied to them. This routine records the voltage it sees on channel zero, counts increments of the multiplexer until the same voltage ($\pm 1/2$ LSB) is found again. Then uses the display for the message "SAME VALUE IN XX ICMX" (ICMX is the increment instruction).

5.2.3 Display - There are four basic patterns which can be displayed: (1) a right diagonal from the lower left corner to the upper right; (2) a left diagonal from the upper left corner to the lower right corner; (3) an angle from the middle to the left edge of the screen to the center then to the top; (4) a box with an "X" and a cross in it. These tests are designed to show inverted bit transfers from accumulator to the deflection registers, malfunctions in the deflection logic and deflection amplifier faults.

5.2.4 INIT3 - This routine is the A/D Converter calibration check. Bits 6-11 of the switch register selects the multiplexer channel, then 1000 octal conversions are made on this channel and stored in a buffer. The first word of the buffer, which is the binary value of the analog voltage is displayed in the upper left of the scope. A horizontal line is now drawn on the scope. A deviation in the amplitude of the horizontal sweep from its point of origin represents a change in the A/D Converter output for a fixed voltage in. A voltage at the converter switching point will give two levels in the horizontal sweep. A noisy converter, or voltage source may give a staircase pattern which indicates excessive noise in the system. The refresh rate of the sweep is controlled by the RC (timing control) clock adjustment on the front panel.

5.2.5 SCOPE - This subroutine call, is placed between each subtest in the instruction section and records the starting address of each subtest as it is being entered. If a scope loop is requested, it will jump to the start of the subtest that the scope loop is requested for.

5.2.6 HALT - This routine printouts an address that tags the failing subtest and the contents of the AC at the time of the failure.

5.3 Program and/or Operator Action

Loading and starting at 200 with all switches down is the start of the IOT instruction test. If an error is detected here, there will be a printout in this section. When an error is detected, and it is necessary to scope on it, place SW9 UP to halt on error, then SW10 UP to loop on it, then SW11 to DELETE printouts.

When errors in the instruction section have been corrected SW2 UP will advance the program to the display test.

SR 0 and 1 select which of the four figures are to be displayed. SR 3 and 4 control the display intensity. SR2 UP allows the program to monitor SW 0, 1, 3 and 4.

When the display is operating correctly SW6 UP will advance the program to the A/D calibration section.

Bits 6 - 11 of the switch register select the multiplexer channel which has a maximum of 27 plus 34, 35, 36 and 37 octal channels.

6. ERRORS

6.1 Error Halt and Description

<u>Halt (PC)</u>	<u>FUNCTION TESTED</u>
0205	DOES START KEY CLEAR SYNC PULSE CHANNEL?
0211	DOES START KEY CLEAR ERROR FLAG?
0221	DOES START KEY CLEAR ALL INTERRUPT ENABLES?
0231	DOES START KEY CLEAR THE RC CLOCK FLAG?
0241	NO RC CLOCK INTERRUPT UP?
0253	(Y) REGISTER EQUAL ZERO (0000)?
0264	(Y) REGISTER EQUAL SEVENS (7777)?
0275	(RADC) INSTRUCTION CLEARS (Y) REGISTER?
0304	CONTINGENCY AND PULSE CHANNEL EQUAL ZERO (0000)?
0313	WILL (ADCV) CONVERT INSTRUCTION SET A/D DONE FLAG?
0324	WILL (RADC) READ BUFFER INSTRUCTION CLEAR A/D DONE FLAG?
0334	WILL (CLER) INSTRUCTION CLEAR A/D DONE FLAG?
0341	DOES (OTEN) INSTRUCTION CLEAR THE AC?
0354	THAT (ICMX) INC MUX DOES NOT SET A/D DONE
0361	THAT (ICMX) DOES NOT CHANGE THE AC
0365	THAT (ICMX) DOES NOT CHANGE THE AC
0375	(Y) REGISTER EQUAL ZERO (0000)
0410	(Y) REGISTER EQUAL SEVENS (7777)

6.1.1 Module Callout For The AX08 - Program cannot be loaded due to bits in AC,

- if bits 0 to 5 B07, B05
- if bits 6 to 11 B08, B06

<u>Address</u>	<u>Good (AC)</u>	<u>Bad (AC)</u>	<u>Corrective Action</u>
0205	0000	XX00	A13, A15
↓	0000	Bit 0	C08
	0000	Bit 1, 2	C07
	0000	Bit 3	C06
0211	0000	NA	D19, C13, A13, A15

<u>Address</u>	<u>Good (AC)</u>	<u>Bad (AC)</u>	<u>Corrective Action</u>
0221	0000	NA	C13, B14, C03, B13
0231	0000	NA	C12, C13
0241	0000	NA	C13
0253	0000	XXXX	D17, D15, A11, A10, A9, D19
↓	0000	7400	C20, B7
	0000	0200	C20, B7
	0000	0100	C19, B7
	0000	0040	C19, B8
	0000	Bit 7 or 8	C18, B8
	0000	Bit 9 or 10	C17, B8
	0000	0001	C16, B8
	0000	7777	A17, B14, A10, A8
	0000	0000	A17, A13, A12, A16, B16
	7777	0377	C20, C14, B02, B07
0264	0000	7777	C20, C14, B02, B07
↓	7777	7577	C20, B07, C14, B02
	7777	7677	C19, B07, C14, B03
	7777	7737	C19, B08, C14, B03
	7777	7757	C18, B08, C15, B03
	7777	7767	C18, B08, C15, B04, B03
	7777	7773	C17, B08, C15, B04
	7777	7775	C17, B08, C15, B04
	7777	7776	C16, B08, C15, B04
	0000	0XXX	B12
	0000	Bit 10, 11	B11
(0275)	0000	Bit 7, 8, 9	B10
↓	0000	Bit 4, 5, 6	B9
	0000	NA	B14, B17, A08, D23, D22, D15, D21, B21, B18, B19, B20, A11, A12, A13, D13
	0000	NA	B17, D18
	0000	NA	A11, A14, D18
	0000	7777	A14, A11
	NA	NA	A7
	0000	XXXX	B14, A10
	7777	XXXX	B14, A10
0304	0000	XXXX	B14, A10
0313	0000	XXXX	B14, A10
0324	NA	NA	B17, D18
0334	NA	NA	A11, A14, D18
0341	0000	7777	A14, A11
0354	NA	NA	A7
0361	0000	XXXX	B14, A10
0365	7777	XXXX	B14, A10

<u>Address</u>	<u>Good (AC)</u>	<u>Bad (AC)</u>	<u>Corrective Action</u>
0375	0000	7777	A08, A07
0410	0000	7777	A08, A07
0421	7777	0000	A08, A07
0430	0000	0000	A08
0440	NA	NA	A08, A10
0445	NA	NA	C13, C19, A18, D15, B13, A12
0463	0000	0000	D19, D18
0470	0000	0000	D19, D18
0475	0000	0000	D13, D15, D18
0504	0000	0000	D18, A14, A11
0520	0000	0000	D18, D19
0533	0000	0000	B14, C03
0550	0000	0000	C13, C03
0570	0000	0000	B14, C03, A15
0612	0000	0000	C13, C03
0625	NA	NA	A08
0635	NA	NA	A07
0637	NA	NA	A07
0646	NA	NA	A07
0656	NA	NA	A07
0660	NA	NA	A07
0670	0000	0000	C4, F28, C10, A16, C12, C13, B16
0702	0000	0000	B16, D17
0712	0000	0000	C12, A14
0725	0000	0000	C02, C13
0743	0000	0000	B16, D17
0754	0000	0000	A15, C01
0764	0000	0000	A16
0770	0000	0000	A16, C9, C8
1014	0000	0000	C13
1025	0000	0000	C11, C12, C13, B13
1040	0000	0000	B12, C12
1053	0000	0000	C11

<u>Address</u>	<u>Good (AC)</u>	<u>Bad (AC)</u>	<u>Corrective Action</u>
0066	0000	0000	C11
1102	0000	0000	C13, C02
1112	4000	0000	C08, A13, A12, B01, B05
1122	0000	0000	C13
1137	NA	NA	D17, D15/CH 34 SHOULD HAVE NON-ZERO VOLTAGE APPLIED
1151	XXXX	7777	A16, B16/CH 34 SHOULD HAVE NON-ZERO VOLTAGE
1160	7777	NA	A16, B16
1167	0000	0000	D17, D15
1241	0000		F28
1303	0000		F28
1324			C8, C9, C10
1461	0000	0000	C14, B13, A24/EXTERNAL INTERRUPT
1503	0000	X000	E27, E28, C07/EXTERNAL STIMULUS
	0000	0X00	E26, 06/EXTERNAL STIMULUS
1524	NA	NA	D17

6.1.2 Display section of list

- if no Intensity, A21, A18, A16, A24
- cannot vary Intensity, A19, A20, B16, A24
- if Alphanumeric is displayed as a series of dots, Intensity duration is too long or not applied to scope.

0421 (RADC + 10) INSTRUCTION CLEARS THE (Y) REGISTER
 0430 WILL (ADCV + 10) CONVERT INSTRUCTION SET A/D DONE FLAG?
 0440 WILL (RADC + 10) READ BUFFER INSTRUCTION CLEAR A/D DONE FLAG?
 0445 WILL TWO (ADCV) RAISE ERROR CONDITION?
 0463 DOES (CLER), CLEAR THE ADCIP "FF"?
 0470 TEST FOR NO ADC TIMING ERROR
 0475 SET ERROR CONDITION
 0504 DOES (CLER) CLEAR ERROR FLAG?
 0520 DOES (CLER) CLEAR THE (ADCIF) "FF"?
 0533 TEST A/D DONE INTERRUPT ENABLE
 0550 TEST ERROR INTERRUPT ENABLE
 0570 TEST A/D DONE INTERRUPT ENABLE
 0612 TEST ERROR INTERRUPT ENABLE
 0625 TEST A/D DONE WITH MICRO PROGRAMMED IOT'S

0635	WILL (ICMX AND ADCV) SET A/D DONE?
0637	WILL MICRO PROGRAMMED IOT RAISE ERROR FLAG"
0646	SET A/D DONE WITH (ACMX, RADCV, AND ADCV)
0656	SET A/D DONE WITH MICRO PROGRAMMED IOT'S
0660	RAISE ERROR CONDITION WITH MICRO PROGRAMMED IOT'S
0670	SET RC CLOCK FLAG
0702	RAISE ERROR WITH RC CLOCK
0712	CLEAR RC CLOCK FLAG
0725	RAISE INTERRUPT WITH RC CLOCK
0743	SET A/D DONE FLAG WITH RC CLOCK
0754	CLEAR RC CLOCK AND TEST FOR NO SKIP
0764	RC CLOCK (SLOW) SET TOO SOON
0770	RC CLOCK (SLOW) NOT SET
1014	RAISE AN INTERRUPT WITH RC CLOCK
1025	RAISE CRYSTAL CLOCK FLAG
1040	CLEAR CRYSTAL CLOCK FLAG
1053	TEST CRYSTAL CLOCK DOES NOT SET FLAG TOO SOON
1066	TEST CRYSTAL CLOCK DOES NOT SET FLAG TOO SLOW
1102	INTERRUPT ON CRYSTAL CLOCK
1112	SET CHANNEL FLAG
1122	TEST CRYSTAL INTERRUPT ENABLE
1137	(DYC) INHIBITED BY CONVERT COMMAND
1151	(DYL) INHIBITED BY CONVERT COMMAND
1160	(DXL) INHIBITED BY CONVERT COMMAND
1167	(DXC) INHIBITED BY CONVERT COMMAND
1241	TEST RC CLOCK STABILITY AT ONE EIGHTH SPEED
1303	TEST RC CLOCK STABILITY NORMAL SPEED
1324	TEST RC CLOCK FAST/SLOW RATIO
1461	TEST FOR ABSENCE OF EXTERNAL INTERRUPT
1503	TEST FOR ABSENCE OF EXTERNAL STIMULUS
1524	TEST THAT RC CLOCK DOES NOT INHIBIT ADCV

6.2 Error Recovery

Depress Continue or Restart at 200

7. RESTRICTIONS

Starting restrictions always starts at 200. There are no operation restrictions.

8. MISCELLANEOUS

Execution time for section 1 is about 1 sec.

9. PROGRAM DESCRIPTION

The three sections of this program are designed to be an increment test of the system. The first section which is a test of hardware's logic, checks that start key initiated the system correctly.

The following are tested automatically by the computer: flags, interrupts, crystal and RC clocks, Y register, timing error conditions, external registers, and other integrated logic functions that are associated with the display and analog digital converter.

The second section uses the display logic and the scope to validate the display. The switch register is used in the test to control the generation of intensity and pattern selection. At this stage of the test, the operator must determine if the display responds correctly in relation to the switch register.

The third section of the test is a check on the calibration and repetition of the A/D Converter and its respective multiplexer channel. The switch register selects the multiplexer channels; conversions on that channel are made and stored into a buffer. The buffer is then displayed on the scope. The first buffer voltage is displayed as an octal value; a deviation in horizontal sweep indicates noise in the system, measuring monotonicity, repeatability and drift. Using this routine is a subjective check on the A/D portions and the environment in which the system is installed; the quality of analog input must be considered in evaluating this display.

11. FLOW CHART

Flow of AX08 Diagnostic

With all switches down test will stay in this section

11=1 Inhibit printout

10=1 Scope Loop

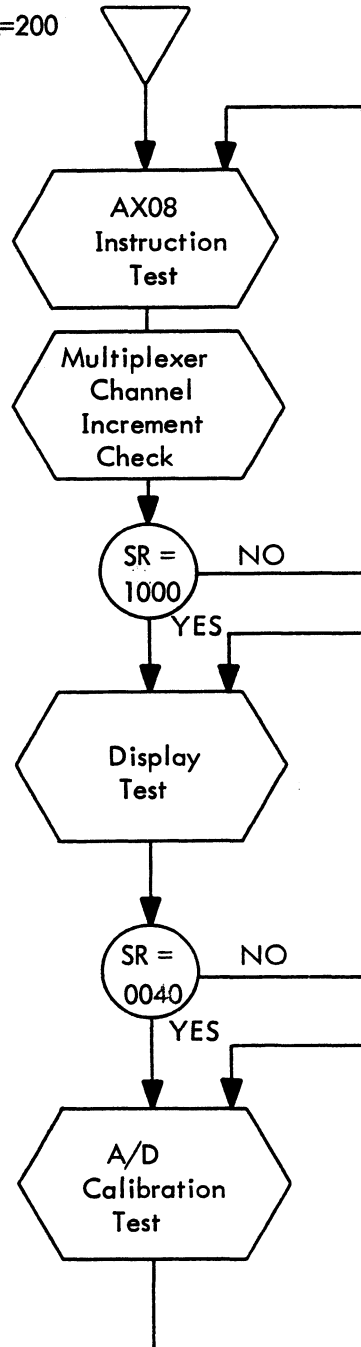
9=1 Halt on Error

Switch Register Control Sweep and Intensity

<u>Patterns</u>		<u>Intensity</u>	
<u>0</u>	<u>1</u>	<u>3</u>	<u>4</u>
0	0 = 	1	0 = Bright
0	1 = 	0	0 = Normal
1	1 = 	0	1 = Dim
1	0 = 		

Test indicates Binary value of Analog Converted Voltage.
Amplitude changes in sweep indicate multi-state condition.
Switch register 6 to 11 select multiplexer channel.

SA=200



/INSTRUCTIONS FOR THE AX08

/

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/

0301	0XC=6301	/CLEAR X REGISTER
0302	0XL=6302	/LOAD X REGISTER (1) SIDE
0304	0IS=6304	/DISPLAY
0311	0YC=6311	/CLEAR Y REGISTER
0312	0YL=6312	/LOAD Y REGISTER (1) SIDE
0321	0XXK=6321	/SKIP ON CRYSTAL CLOCK
0322	0XER=6322	/SKIP ON ADC TIMING ERROR
0324	0S- =6324	/SET BRIGHTNESS
0331	0XIN=6331	/OR EXTERNAL SENSE REGISTER INTO AC
0332	0XAD=6332	/SKIP ON A/D DONE
0334	0XCL=6334	/CLEAR SENSE BIT WITH ONE IN AC
0341	0XHK=6341	/SKIP ON RC TIMING CLOCK
0342	0XIN=6342	/ZERO IN AC INHIBIT INTERRUPT REGISTER
0344	0XEN=6344	/ONES IN AC ENABLE INTERRUPT
0351	0LER=6351	/CLEAR ADC TIMING ERROR CONDITION
0352	0LAK=6352	/CLEAR CRYSTAL CLOCK FLAG
0354	0L-K=6354	/CLEAR RC CLOCK AND FLAG
0361	0CX=6361	/INCREMENT MULTIPLEXER
0362	0RAC=6362	/CLEAR AC, READ A/D BUFFER
0364	0ACV=6364	/START CONVERSION
0371	0CX=6371	/JAM SET MULTIPLEXER
7344	MI=CLA CLL CMA RAL	
0020	*20	
0021	M/400, -/400	
0022	K/400, /400	
0023	KM040, -0040	
0024	K0400, 0400	
0025	K00777, -0777	
0026	SL1, 0	
0027	FL1, 0	
0030	M0377, -0377	
0031	K0034, 0034	
0032	K0100, 0100	
0033	K1000, 1000	
0034	K0002, 0002	
0035	K2002, 2002	
0036	K3002, 3002	
0037	K0004, 0004	
0040	C0070, 0070	
0041	K0200, 0200	
0042	K1004, 1004	
0043	K0340, 0340	
0044	K4002, 4002	
0045	K0015, 0015	
0046	K7700, 7700	
0047	K0077, 0077	
0050	K01000, -1000	
0051	KM020, -20	
0052	K1002, 1002	
0053	K776, 776	

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0003 1111
0004 4200
0005 3500
0006 3755
0007 0200
0008 0213

K1111, 1111
MSG, MSG
CHUSC1, CHUSC
AUSTBL, USCTBL
MAXNUM, 2
K2103, 0003

0061	7772	KM0000, -6
0062	0000	CIEMA, 0
0063	0000	CIEMB, 0
0064	0000	CIEMC, 0
0065	0000	CCONTA, 0
0066	0000	CCONTB, 0
0067	0000	CCONTC, 0
0070	0000	HAD, 0
0071	4000	INLL, 4000
0072	0000	LOOP, 0
0073	0000	0
0074	0000	XER, 0
0075	0000	TEMP01, 0
0076	0000	TEMP02, 0
0077	0000	TEMP03, 0
0100	0000	TEMP04, 0
0101	0000	TEMP05, 0
0102	0000	TEMP06, 0
0103	0000	MSKHI, 0
0104	0000	CHXL, 0
0105	0000	CHYL, 0
	4454	LOH=JMS 1 MSSG
0106	0000	STALL, 0
0107	0040	K0040, 0040
0110	2400	XINITS, INITS
0111	4223	XSCOPE, SCOPEA
	4511	SCOPE=JMS 1 XSCOPE
0112	4261	XERADD, ERAUD
0113	4320	XIYPE, MESSAGE
0114	4235	XIPRIN, IPRINT
0115	1642	XSIXTY, SIXTY
0116	4252	XCRLF, CRLF
0117	1500	XTEXTB, TEXTB
0120	0000	AC, 0
0121	2000	XSWTCH, SWICH
	7000	OPR=7000
	7402	XX=7402
	4512	HALT=JMS 1 XERADD

0122 2147
 0123 2237
 0124 2256
 0125 2160
 0126 2200
 0127 2212
 0130 2224

STORA, STORS
 DLOLA, DLOLS
 DROLA, DROLS
 DVLTA, DVLTS
 DVLBA, DVLBS
 DHLRA, DHLRS
 DHLLA, DHLLS

0131 0500
 0132 6324
 0133 0377
 0134 0000
 0135 0000
 0136 0400
 0137 0777
 0140 0002
 0141 7400
 0142 7600
 0145 0000

K0000, 000
 K0324, 6324
 K377, 377
 XPOSR, 0
 YPOSR, 0
 K400, 400
 K777, 777
 P2, 2
 M400, -400
 M200, -200
 COUNT, 0

4522
 4525
 4526
 4527
 4530
 4523
 4524

STOR=JMS I STORA
 DVL I=JMS I DVLTA
 DVLB=JMS I DVLBA
 DHLR=JMS I DHLRA
 DHLL=JMS I DHLLA
 DLOL=JMS I DLOLA
 DRDL=JMS I DRDLA
 /SR
 /01

/00=LEFT DIAGONAL 0=REPEAT
 /01=RIGHT DIAGONAL
 /10=ANGLE 1=NEW TEST
 /11=XBOX
 *. 17/+1

BRIGHTNESS

0200

/START KEY SHOULD CLEAR SYNC PULSE CHANNEL

0200	7200	BEGIN, CLA	
0201	6331	XBIN	/READ PULSE REGISTER
0202	0021	AND K7400	/BIT 0 TO 3
0203	7440	SEA	
0204	4512	HALT	/NOT CLEARED BY START KEY
0205	4511	SCOPE	
0206	6322	/DOES START KEY CLEAR ERROR FLAG?	
0207	7410	SKN	/SKIP ON ERROR FLAG
0210	4512	HALT	/ERROR FLAG SKIPPED
0211	4511	SCOPE	
0212	1223	/TEST FOR NO INTERRUPT /START KEY SHOULD CLEAR ALL INTERRUPT ENABLES	
0213	3001	IAU .+11	
0214	1224	OCA 1	
0215	3002	IAU .+10	
0216	0001	OCA 2	
0217	7410	ION	
0220	4512	SKP	
0221	0002	HALT	/INTERRUPT UP, START KEY
0222	5225	IOF	/SHOULD CLEAR ALL INTERRUPT ENABLES
0223	5402	JMP .+3	
0224	0220	JMP 1 2	/GO TO MEMORY LOCATION ONE
0225	4511	.-4	/GO TO MEMORY LOCATION TWO
0226	6341	SCOPE	
0227	7410	/DOES START KEY CLEAR RC CLOCK FLAG	
0230	4512	SKN	/SKIP ON RC CLOCK
0231	4511	SKP	
0232	1032	HALT	/START KEY DID NOT CLEAR FLAG
0233	6340	SCOPE	
0234	1243	/TEST FOR NO RC CLOCK INTERRUPT	
0235	5002	IAU K1000	/SET RC INTERRUPT ENABLE
0236	0001	ATEN OTEN	
0237	7410	IAU .+7	
0240	4512	OCA 2	
0241	0002	ION	/THERE SHOULD BE NO INTERRUPT
0242	7410	SKP	
0243	0240	HALT	/RC CLOCK INTERRUPT UP
0244	4511	IOF	
0245	6311	SKP	
0246	6362	.-3	
0247	7650	SCOPE	
0250	5253	/TEST READ BUFFER (Y REGISTER) AC=ZERO	
0251	6362	LOOPA, JYC	/CLEAR Y REGISTER
0252	4512	RAUC	/READ BUFFER
0253	4511	SNA CLA	/TEST
		JMP .+3	/GOOD
		RAUC	/BAD
		HALT	/AC SHOULD EQUAL 0000
		SCOPE	
0254	7240	/TEST READ BUFFER (Y REGISTER) AC=SEVEN	
0255	6311	CLA CMA	/SET AC TO SEVENS
		JYC	/CLEAR Y REGISTER

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0200 6362
0201 7450
0202 5264
0203 7240
0204 6362
0205 4512
0206 4511

RAUC /READ BUFFER
SNA /TEST
JMP ,+4 /GOOD
CLA CMA /BAD
RAUC /AC SHOULD EQUAL 0000
HALT /BAD AC CLEAR PULSE
SCOPE


```

0265 7240      /TEST READ BUFFER (Y REGISTER)
0266 6312      CLA CMA          /AC EQUAL SEVENS
0267 6362      DYL            /SET Y REGISTER
0270 7201      RAUC          /READ BUFFER
0271 7650      IAC
0272 5275      SNA CLA        /TEST
0273 6362      JMP ,+3        /GOOD
0274 4512      RAUC          /BAD
0275 4511      HALT          /AC SHOULD EQUAL 7777
0276 7240      SCOPE
0277 6334      /TEST CONTINGENCY AND PULSE CHANNEL FOR ZERO
0300 7200      CLA CMA        /AC=SEVENS
0301 6331      XRCL          /CLEAR CHANNEL
0302 7440      CLA
0303 4512      XRIN          /READ CHANNELS
0304 4511      SZA          /TEST FOR ZERO
0305 4511      HALT          /FAILED TO CLEAR
0306 4511      SCOPE
0307 6364      /WILL CONVERT INSTRUCTION SET THE FLAG?
0308 3106      ADCV          /START
0309 2106      UCA STALL
0310 5307      ISZ STALL      /TIME FOR FLAG TO SET
0311 6332      JMP , -1
0312 4512      SKAU          /SKIP ON DONE FLAG
0313 6362      HALT          /DONE FLAG NOT SET OR SKIP FAILED
0314 4511      RAUC          /CLEAR FOR SCOPE
0315 4511      SCOPE
0316 6364      /TEST THAT (RADU) CLEARS DONE FLAG
0317 6332      ADCV          /START CONVERT TO SET FLAG
0318 5316      SKAU
0319 6362      JMP , -1      /WAIT FOR FLAG
0320 6362      RAUC          /CLEAR FLAG
0321 6332      SKAU          /IS FLAG CLEARED
0322 7410      SKP          /YES
0323 4512      HALT          /NO, FLAG IS SET OR SKIP FAILED
0324 4511      SCOPE
0325 6364      /TEST THAT (CLER) CLEARS THE DONE FLAG
0326 6332      ADCV          /START CONVERT TO SET FLAG
0327 5325      SKAU          /SKIP ON DONE
0328 6351      JMP , -1      /WAIT FOR FLAG
0329 6351      CLER          /CLEAR DONE FLAG
0330 6332      SKAU          /TEST THAT FLAG IS CLEARED
0331 7410      SKP
0332 4512      HALT          /CLER DID NOT CLEAR DONE FLAG
0333 4511      SCOPE
0334 7240      /TEST THAT (OTEN) CLEARS THE AC
0335 6344      CLA CMA        /AC=SEVENS
0336 7440      OTEN          /CLEAR AC
0337 4512      SZA          /TEST
0338 4511      HALT          /OTEN DID NOT CLEAR AC
0339 4511      SCOPE
0340 6342      ZTEN          /CLEAR ALL INTERRUPT ENABLES
0341 2106      /WILL INC MUX SET DONE FLAG
0342 2106      ISZ STALL

```

0344	5343	JMP .-1	
0345	6351	CLER	
0346	0361	ICMX	/INCREMENT MUX
0347	2106	ISZ STALL	
0350	5347	JMP .-1	
0351	0352	SKAU	/SKIP ON FLAG
0352	7410	SKP	
0353	4512	HALT	/ICMX SET DONE FLAG
0354	0302	RAUC	
0355	4511	SCOPE	

```

/WILL MULTIPLEXER INSTRUCTION CHANGE AC
0356 6361      ICMX      /INCREMENT MUX
0357 7440      SZA
0358 4512      HALT      /ICMX, SET AC
0359 7240      CLA CMA
0360 6361      ICMX      /INCREMENT MUX
0361 7450      SNA
0362 4512      HALT      /ICMX+10, SET AC
0363 6362      RAUC
0364 4511      SCOPE
/ROUTINES USING IOT 637X
/TEST READ BUFFER (Y REGISTER) AC=ZERO
0367 6311      DYC      /CLEAR Y REGISTER
0368 6372      RAUC+10   /READ BUFFER
0369 7650      SNA CLA   /TEST
0370 5375      JMP ,+3    /GOOD
0371 6372      RAUC+10   /BAD
0372 4512      HALT      /AC SHOULD EQUAL 0000
0373 4511      SCOPE
0374 5777      JMP I ,+1
0375 177+1
/TEST READ BUFFER (Y REGISTER) AC=SEVEN
0400 7240      CLA CMA   /SET AC TO SEVENS
0401 6311      DYC      /CLEAR Y REGISTER
0402 6372      RAUC+10   /READ BUFFER
0403 7450      SNA       /TEST
0404 5210      JMP ,+4    /GOOD
0405 7240      CLA CMA   /BAD
0406 6372      RAUC+10   /AC SHOULD EQUAL 0000
0407 4512      HALT      /BAD AC CLEAR PULSE
0408 4511      SCOPE
/TEST READ BUFFER (Y REGISTER)
0411 7240      CLA CMA   /AC EQUAL SEVENS
0412 6312      DYL      /SET Y REGISTER
0413 6372      RAUC+10   /READ BUFFER
0414 7001      IAC
0415 7650      SNA CLA   /TEST
0416 5221      JMP ,+3    /GOOD
0417 6372      RAUC+10   /BAD
0418 4512      HALT      /AC SHOULD EQUAL 7777
0419 4511      SCOPE
/WILL CONVERT INSTRUCTION SETS THE FLAG
0422 6374      ADCV+10   /START
0423 3106      DCA STALL
0424 2106      ISZ STALL /TIME FOR FLAG TO SET
0425 5224      JMP , -1
0426 6332      SKAD      /SKIP ON DONE FLAG
0427 4512      HALT      /DONE FLAG NOT SET OR SKIP FAILED
0428 4511      SCOPE

```



```

/TEST THAT (RADC) CLEARS DONE FLAG
0431 6374      ADCV+10      /START CONVERT TO SET FLAG
0432 6332      SKAU
0433 5232      JMP , -1      /WAIT FOR FLAG
0434 6372      RADC+10      /CLEAR FLAG
0435 6332      SKAU          /IS FLAG CLEARED
0436 7410      SKP          /YES
0437 4512      HALT          /NO, FLAG IS SET OR SKIP FAILED
0440 4511      SCOPE

/GENERATE TIMING ERROR CONDITION
0441 6374      ADCV+10      /CONVERT
0442 6374      ADCV+10      /TWO ADCV SET ERROR CONDITION
0443 6322      SKLR        /SKIP ON ERROR
0444 4512      HALT        /ERROR NOT UP
0445 6332      SKAU
0446 5245      JMP , -1
0447 4511      SCOPE

/DOES (CLER) CLEAR ADCIP "FF"
0450 6374      ADCV+10
0451 6374      ADCV+10      /SET ERROR CONDITION
0452 6332      SKAU
0453 5252      JMP , -1      /WAIT FOR FLAG
0454 6351      CLER        /CLEAR ADCIP "FF"
0455 6374      ADCV+10      /CONVERT
0456 6332      SKAU        /SKIP ON FLAG
0457 5256      JMP , -1      /WAIT FOR FLAG
0460 6322      SKLR        /SKIP ON ERROR
0461 7410      SKP          /NO ERROR
0462 4512      HALT        /((CLER) DID NOT CLEAR ADCIP FF
0463 4511      SCOPE

```

```

/TEST FOR ADC TIMING ERROR
0404 6351      CLER      /CLEAR ERROR CONDITION
0405 6322      SKER      /SKIP ON ERROR
0406 7410      SKP
0407 4512      HALT      /ERROR CONDITION UP
0408 4511      SCOPE

/GENERATE TIMING ERROR CONDITION
0411 6364      ADCV      /CONVERT
0412 6364      ADCV      /TWOADCV SET ERROR CONDITION
0413 6322      SKER      /SKIP ON ERROR
0414 4512      HALT      /ERROR NOT UP
0415 6332      SKAD
0416 5215      JMP , -1
0417 4511      SCOPE

/DOES (CLER) CLEAR ERROR FLAG
0500 6351      CLER      /CLEAR ERROR
0501 6322      SKER      /SKIP ON ERROR
0502 7410      SKP      /NO ERROR
0503 4512      HALT      /ERROR FLAG NOT CLEARED
0504 4511      SCOPE

/DOES (CLER) CLEAR ADCIP "FF"
0505 6364      ADCV
0506 6364      ADCV      /SET ERROR CONDITION
0507 6332      SKAD
0510 5307      JMP , -1      /WAIT FOR FLAG
0511 6351      CLER      /CLEAR ADCIP "FF"
0512 6364      ADCV      /CONVERT
0513 6332      SKAD      /SKIP ON FLAG
0514 5313      JMP , -1      /WAIT FOR FLAG
0515 6322      SKER      /SKIP ON ERROR
0516 7410      SKP      /NO ERROR
0517 4512      HALT      /(CLER) DID NOT CLEAR ADCIP FF
0520 4511      SCOPE

```

/SET DONE AND TEST FOR INTERRUPT WITH NO INTERRUPT ENABLES

0021	6364	ADCV	/START CONVERT
0022	6332	SKAU	
0023	5322	JMP , -1	/WAIT FOR DONE FLAG
0024	7200	CLA	
0025	6342	ETEN	/INHIBIT INTERRUPTS
0026	1332	IAU , +7	
0027	3002	JCA 2	
0030	6001	ION	
0031	7410	SKP	
0032	4512	HALT	/INTERRUPT UP
0033	6002	IOF	
0034	7410	SKP	
0035	0532	. -3	
0036	4511	SCOPE	

/SET ERRR FLAG AND TEST FOR INTERRUPT WITH NO INTERRUPT ENABLE

0037	6364	ADCV	
0040	6364	ADCV	/GENERATE ERROR FLAG
0041	6332	SKAU	
0042	5341	JMP , -1	
0043	1332	IAU , +7	
0044	3002	JCA 2	
0045	6001	ION	
0046	7410	SKP	
0047	4512	HALT	/INTERRUPT UP
0050	6002	IOF	
0051	7410	SKP	
0052	0547	. -3	
0053	4511	SCOPE	

/WILL DONE FLAG RAISE AN INTERRUPT

0054	7200	CLA	
0055	1031	IAU K0100	
0056	6346	ETEN OTEN	/ENABLE ADC INTERRUPT
0057	6364	ADCV	/START CONVERT
0060	6332	SKAU	
0061	5360	JMP , -1	/WAIT FOR FLAG
0062	1371	IAU , +7	
0063	3002	JCA 2	
0064	6001	ION	
0065	7000	NOP	
0066	6002	IOF	
0067	4512	HALT	/DONE FLAG DID NOT RAISE INTERRUPT
0070	7410	SKP	
0071	0572	. +1	
0072	4511	SCOPE	
0073	5774	JMP I , +1	
0074	0600	. 1/7+1	

```

/Will TIMING ERROR RAISE AN INTERRUPT
0575 7200      CLA
0576 1A40      IAD K0200
0577 0546      ZTEN UTEN
0578 0504      ADCV
0579 0504      ADCV
0580 0532      SKAU
0581 5202      JMP .-1
0582 1210      IAD .+7
0583 0502      JCA 2
0584 0801      ION
0585 7A00      NOP
0586 0502      IOF
0587 4512      HALT      /NO INTERRUPT
0588 7410      SKP
0589 0512      .-1
0590 0501      CLER
0591 4511      SCOPE

/MICRO PROGRAMED IOT
0515 0502      RAUC      /CLEAR DONE
0516 7240      CLA CMA
0517 0507      ICMX RAUC ADCV /INC MUX, READ BUFFER, CONVERT
0518 2100      ISZ STALL
0519 5221      JMP .-1
0520 0532      SKAU      /SKIP ON      DONE
0521 4512      HALT      /DONE FLAG NOT SET
0522 4511      SCOPE

/SET TIMING ERROR FLAG
0526 0501      CLER
0527 0507      ICMX ADCV
0528 0507      ICMX ADCV
0529 2100      ISZ STALL
0530 5231      JMP .-1
0531 0532      SKAU
0532 4512      HALT      /DONE FLAG NOT SET
0533 0522      SKER
0534 4512      HALT      /NO TIMING ERROR FLAG
0535 4511      SCOPE
0536 0502      RAUC
0537 0507      ACMX RAUC ADCV      /SET MUX, READ BUFFER, CONVERT
0538 2100      ISZ STALL
0539 5242      JMP .-1
0540 0532      SKAU
0541 4512      HALT      /DONE FLAG NOT SET
0542 4511      SCOPE

```

/SET TIMING ERROR FLAG

0547	6351	CLER	
0550	6377	ACMX RADC ADCV	/SET MUX, READ BUFFER, CONVERT
0551	6375	ACMX ADCV	
0552	2106	ISZ STALL	
0553	5252	JMP, -1	
0554	6332	SKAD	/SKIP ON DONE
0555	4512	HALT	/DONE NO SET
0556	6322	SKER	/SKIP ON ERROR
0557	4512	HALT	/ERROR SHOULD BE UP
0560	4511	SCOPE	

/TEST USING RC CLOCK CONVERTING + INTERRUPT

/TEST RC CLOCK

/ENABLE RC CLOCK AND SKIP ON FLAG

0501	7200	CLA	
0502	1033	IAU K2002	
0503	6346	ZTEN OTEN	/TURN ON RC CLOCK
0504	2100	ISE STALL	
0505	0504	JMP .-1	
0506	6341	SKKK	/RC CLOCK FLAG SHOULD BE SET
0507	4012	HALT	/DID NOT SKIP ON RC CLOCK FLAG
0508	6354	CLRK	/CL RC CLOCK AND FLAG
0509	4011	SCOPE	

/TEST FOR ADC TIMING ERROR

0512	6351	CLRK	
0513	7200	CLA	
0514	1034	IAU K2002	
0515	6346	ZTEN OTEN	
0516	2100	ISE STALL	
0517	0516	JMP .-1	
0518	6322	SKKK	/SKIP ON ADC TIMING ERROR
0519	4012	HALT	/ERROR SHOULD BE SET
0520	4011	SCOPE	/RC TIME ADJ SET FOR MIN

/TEST CLEAR RC CLOCK INSTRUCTION

0703	6354	CLRK	/CLEAR RC CLOCK AND FLAG
0704	6341	SKKK	
0705	0304	JMP .-1	/SYNC ON CLOCK
0706	6354	CLRK	
0707	6341	SKKK	/SKIP ON RC CLOCK
0710	7410	SKP	
0711	4012	HALT	/CLEAR DID NOT WORK
0712	4011	SCOPE	

/WILL RC CLOCK RAISE AN INTERRUPT

0713	6341	SKKK	
0714	0313	JMP .-1	/WAIT FOR RC CLOCK FLAG
0715	1035	IAU K3002	
0716	6346	ZTEN OTEN	/RC CLOCK INTERRUPT ENABLE ON
0717	1326	IAU .+7	
0720	0002	OCA 2	
0721	0001	ION	
0722	7000	VOP	/SHOULD INTERRUPT HERE
0723	6002	IOP	
0724	4012	HALT	/NO INTERRUPT
0725	7410	SKP	
0726	0725	.-1	
0727	6354	CLRK	
0730	4011	SCOPE	

/WILL RC CLOCK SET ADC DONE FLAG

0/31	6354	CLRK	
0/32	6341	SKRK	
0/33	5332	JMP .-1	
0/34	6354	CLRK	
0/35	6341	SKRK	/SKIP ON RC FLAG
0/36	5332	JMP .-1	/WAIT FOR RC FLAG
0/37	2106	ISE STALL	
0/40	5337	JMP .-1	/ADC DONE SHOULD BE SET
0/41	6332	SKAD	/SKIP ON ADC DONE
0/42	4512	HALT	/ADC DONE NOT SET
0/43	4511	SCOPE	

/CLEAR RC CLOCK AND TEST FOR NO SKIP

0/44	7200	CLA	
0/45	6342	ZTEN	/TURN CLOCK OFF
0/46	6354	CLRK	/CL CLOCK FLAG
0/47	2106	ISE STALL	
0/50	5347	JMP .-1	
0/51	6341	SKRK	/SKIP ON RC CLOCK
0/52	7410	SKP	
0/53	4512	HALT	/CLOCK SHOULD BE OFF
0/54	4511	SCOPE	

/TEST RC CLOCK (ONE EIGHT SPEED)

0/55	7200	CLA	
0/56	1043	TAU K4002	/SLOW SPEED RC CLOCK ENABLE
0/57	6346	ZTEN OTEN	/CLEAR RC CLOCK FLAG
0/60	6354	CLRK	/SKIP ON RC FLAG
0/61	6341	SKRK	
0/62	7410	SKP	
0/63	4512	HALT	/RC FLAG SET TOO SOON
0/64	2106	ISE STALL	
0/65	5364	JMP .-1	/WAIT FOR FLAG TO SET
0/66	6341	SKRK	/SKIP ON RC FLAG
0/67	4512	HALT	/FLAG NOT SET
0/70	4511	SCOPE	
0/71	5772	JMP I .+1	
0/72	1000	. 1/7+1	

*. 177+1

/TEST THAT RC CLOCK GENERATES INTERRUPTS

1000	7000	UPR	
1001	7200	CLA	
1002	1051	TAU K1002	/RC INTERRUPTS AND RC TIMING
1003	6346	ZTEN OTEN	
1004	2106	ISE STALL	
1005	5204	JMP .-1	
1006	1215	TAU .+7	
1007	5002	UCA 2	
1010	6001	ION	
1011	7000	UPR	
1012	6002	IOF	
1013	4512	HALT	/NO INTERRUPT
1014	7410	SKP	
1015	1014	.-1	
1016	6351	CLEN	

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101/ 6354
1020 4511

CLRK
SCOPE

```
1021 3106      /TEST USING CRYSTAL CLOCK TO SET FLAG
1022 2106      UCA STALL      /FLAG SHOULD BE SET 150 USEC
1023 5226      ISZ STALL
1024 4012      JMP ,+3
1025 5230      HALT          /CRYSTAL CLOCK FLAG NOT SET
1026 0321      JMP ,+3
1027 5222      SKXX          /SKIP ON CRYSTAL FLAG
1030 4011      JMP , -5      /NOT SET
                        SCOPE
```

PAUSE


```

/TAPE 2
/CLEAR CRYSTAL CLOCK FLAG WITH "IOT"
1031 0322      CLXX
1032 0321      SKXX
1033 5232      JMP , -1
1034 0352      CLXX
1035 0321      SKXX
1036 7410      SKP
1037 4512      HALT
1040 4511      SCOPE
/TEST THAT FLAG DOES NOT SET TOO SOON
1041 0352      CLXX
1042 0321      SKXX
1043 5242      JMP , -1          /SYNC ON CLOCK
1044 0352      CLXX
1045 2106      ISZ STALL
1046 2106      ISZ STALL
1047 2106      ISZ STALL
1050 0321      SKXX
1051 7410      SKP
1052 4512      HALT          /FLAG SET TOO SOON
1053 4511      SCOPE
/TEST THAT FLAG DOES NOT TAKE TOO LONG TO SET
1054 0352      CLXX
1055 0321      SKXX
1056 5255      JMP , -1          /SYNC ON CLOCK
1057 0352      CLXX          /CLEAR XTAL CLOCK FLAG
1060 1022      IAU KM0043
1061 3106      UCA STALL
1062 2106      ISZ STALL          /150 MICRO SEC
1063 5262      JMP , -1
1064 0321      SKXX          /SKIP ON XTAL CLOCK FLAG
1065 4512      HALT          /XTAL CLOCK FL TAKES TOO LONG
1066 4511      SCOPE
/WILL CRYSTAL CLOCK GENERATE AN INTERRUPT
1067 7200      CLA
1070 1023      IAU KM0400          /ENABLE XTAL CLOCK INTERRUPT
1071 0346      ZTEN OTEN
1072 2106      ISZ STALL          /XTAL CLOCK FLAG SHOULD BE SET
1073 5272      JMP , -1
1074 1303      IAU , +7
1075 3002      UCA 2
1076 6001      ION          /INTERRUPT ON
1077 7000      NOP          /SHOULD INTERRUPT HERE
1100 6002      IOF
1101 4512      HALT          /NO XTAL INTERRUPT UP
1102 7410      SKP
1103 1104      , +1
1104 4511      SCOPE
/SET CHANNEL FLAG
1105 7124      STL RAL          /0001 TO AC
1106 0346      OTEN ZTEN      /SET CHANNEL FLAG
1107 0331      XRIN          /SENSE REGISTER TO AC

```

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1110 7500
1111 4012
1112 7200
1113 4011

SMA
HALT
CLA
SCOPE

/AC SHOULD=4000
/BIT 0 NOT SET

```

1114 6352      /CLEAR XTAL FLAG AND TEST FOR NO INTERRUPT WITH ENABLE UP
1115 1524      CLXX      /CLEAR CRYSTAL CLOCK FLAG
1116 3002      TAD ,+7
1117 6001      DCA 2
1120 7410      ION
1121 4512      SKP
1122 6002      HALT      /INTERRUPT UP
1123 7410      IOP
1124 1121      SKP
1125 4511      , -3
1126 1030      SCUPE
1127 6371      TAD K0034
              ACMX      /MULTIPLEXER CHANNEL 34

/TEST THAT DISPLAY COMMANDS DO NOT NTERFERE WITH A/D CONVERSIONS
/THIS TEST ASSUMES THAT CHANNEL 34 IS SELECTED AND THAT
/AND THAT A FIXED NON ZERO VOLTAGE IS SUPPLIED TO CHANNEL 34
1130 6364      ADCV      /START CONVERSION
1131 7000      NOP
1132 7000      NOP      /CONVERSION IN PROGRESS
1133 6311      OYC      /THIS INSTRUCTION SHOULD BE INHIBITED
1134 6362      RAUC      /IF ZERO (OYC) NOT INHIBITED
1135 7450      SNA      /TEST
1136 4512      HALT      /OYC WAS NOT INHIBITED AND CLEARED CONVERTER
1137 4511      SCOPE

```



```

/TEST DYL IS INHIBITED DURING A/D CONVERSIONS-CH 34 SET TO HALF POSITIVE SCAL
      ADCV          /START CONVERSION
      SKAD
      JMP , -1      /WAIT FOR FLAG
      CLA CMA       /AC=7777
      DYL           /((DYL) SHOULD BE INHIBITED UNTILL RADC
      RADC          /IF AC=7777 DYL NOT INHIBITED
      JAC
      SNA           /TEST AC NOT EQUAL 7777
      HALT          /DYL NOT INHIBITED
      SCOPE         /CHANNEL 34 SHOULD HAVE A NON ZERO VOLTAGE

```

/TEST THAT (DXL) IS INHIBITED DURING A/D CONVERSION

```

1152 7240
1 53 6364
1 54 6302
1 55 6332
1156 7410
1157 4512
1 56 4511
      CLA CMA       /AC=SEVEN
      ADCV          /START CONVERSION
      DXL           /SHOULD BE INHIBITED AN NOT LOAD "X" WITH SEVENS
      SKAD          /IF "X" IS LOADED, DONE WILL SET TOO SOON
      SKP
      HALT          /DONE SET TOO SOON
      SCOPE

```

/TEST THAT DXC IS INHIBITED BETWEEN (ADCV) AND (RADC)

```

1161 6364
1 52 6332
1 53 5362
1164 6301
1165 6332
1 56 4512
1 57 4511
      ADCV          /START CONVERSION
      SKAD
      JMP , -1      /WAIT FOR FLAG
      DXC           /"DXC" SHOULD BE INHIBITED
      SKAD          /IF NOT INHIBITED WILL CLEAR ADC FLAG
      HALT          /DXC CLEARED DONE
      SCOPE

```

/THIS ROUTINE IS FOR SCOPING RELAY ENABLES R1, R2 AND R4

```

1170 1106
1 71 0037
      6346
      2106
      ISZ STALL
      JMP , -4
      JMP I , +1
      , 177+1
1174 5370
1175 5776
1 76 1200
      1200

```

*. 17/+1

/TEST RC CLOCK SLOW (ONE EIGHT SPEED)

1200	7200	RCX,	CLA	
1201	1043		IAU K4002	
1202	6346		ETEN OTEN	
1203	3106		UCA STALL	
1204	6354		CLRK	
1205	6341		SKRK	
1206	5207		JMP .-1	
1207	6354		CLRK	
1210	2106		ISE STALL	
1211	6341		SKRK	
1212	5210		JMP .-2	
1213	1106		IAU STALL	
1214	3027		UCA SL1	
1215	3106		UCA STALL	
1216	6354		CLRK	
1217	6341		SKRK	
1220	5217		JMP .-1	
1221	6354		CLRK	
1222	2106		ISE STALL	
1223	6341		SKRK	
1224	5222		JMP .-2	
1225	1106		IAU STALL	
1226	7041		CLA	
1227	1027		IAU SL1	
1230	7450		SNA	
1231	5241		JMP RCX1	/EQUAL
1232	7040		CMA	
1233	7450		SNA	
1234	5241		JMP RCX1	/MINUS ONE
1235	7001		IAU	
1236	7040		CMA	
1237	7440		SEA	/PLUS ONE
1240	4012		HALT	
1241	4011	RCX1,	SCOPE	

/TEST RC CLOCK FAST

1242	7200		CLA
1243	1043		IAU K0002
1244	6346		ETEN OTEN
1245	3106		UCA STALL
1246	6354		CLRK
1247	6341		SKRK
1250	5247		JMP .-1
1251	6354		CLRK
1252	2106		ISE STALL
1253	6341		SKRK
1254	5252		JMP .-2
1255	1106		IAU STALL
1256	3026		UCA FT1
1257	3106		UCA STALL
1260	6354		CLRK
1261	6341		SKRK
1262	5261		JMP .-1
1263	6354		CLRK
1264	2106		ISE STALL

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1205	6341	SKKK	
1206	5264	JMP , -2	
1207	1106	TAU STALL	
1210	7041	CIA	
1211	1026	TAU FT1	
1212	7450	SNA	
1213	5303	JMP RCX2	/EQUAL
1214	7040	CMA	
1215	7450	SNA	
1216	5303	JMP RCX2	/MINUS ONE
1217	7001	IAC	
1300	7040	CMA	
1301	7440	SZA	/PLUS ONE
1302	4512	HALT	
1303	4511	RCX2, SCUPE	
		/TEST FAST--SLOW RATIO OF RC CLOCK	
1304	7300	CLA CLL	/IF THIS TEST FAILS
1305	1025	TAU SL1	/RATIO OF CLOCK IS
1306	7110	RAR CLL	/NOT 8 TO 1
1307	7110	RAR CLL	
1310	7110	RAR CLL	
1311	7041	CIA	
1312	1026	TAU FT1	
1313	7450	SNA	
1314	5324	JMP RCX3	
1315	7240	CMA	
1316	7450	SNA	
1317	5324	JMP RCX3	
1320	7001	IAC	
1321	7040	CMA	
1322	7440	SZA	
1323	4512	HALT	/RATIO NOT 8 TO 1
1324	4511	RCX3, SCUPE	/THIS IS NOT TRUE
1325	6342	ETEN	/CLEAR ENABLES
1326	6354	CLRK	/CLEAR RC FLAG
1327	5730	JMP I , +1	/USES DISPLAY TO CHECK ICMX
1330	1400	CHIST	/INSTRUCTION

```

1331 4340      /SELECT MUX WITH SWITCHS, DISPLAY CONVERTED VALUE IN AC
CON3,  JMS CON1
1332 4353      JMS CON2
1333 5351      JMP ,-2
1334 4340      JMS CON1      /USE ACMX, RADCV, ADCV
1335 5354      JMP ,-1
1336 4353      JMS CON2      /USE ICMX RADXC ADCV
1337 5356      JMP ,-1

/DISPLAY CONVERTER VALUE IN AC, SW REGISTER 7-11 SELECT MUX
CON1,  NOP
      LAS
      ACMX ADCV
      RAUC
      SKAU
      JMP ,-1
      ISZ STALL
      JMP ,-1
      ISZ STALL
      JMP ,-1
      JMP I CON1
CON2,  NOP
      LAS
      IAU K777/      /MINUS ONE
      ACMX
      ICMX
      ADCV
      RAUC
      SKAU
      JMP ,-1
      ISZ STALL
      JMP ,-1
      ISZ STALL
      JMP ,-1
      JMP I CON2
      JMP I ,+1
      . 1/7+1
1340 7000
1341 7504
1342 6575
1343 6362
1344 6352
1345 5344
1346 2106
1347 5346
1350 2106
1351 5350
1352 5740
1353 7000
1354 7604
1355 1053
1356 6371
1357 6361
1360 6364
1361 6362
1362 6352
1363 5362
1364 2106
1365 5364
1366 2106
1367 5366
1370 5753
1371 5772
1372 1400

```

```

1400
1400 7200
1401 3057
1402 6371
1403 6364
1404 6332
1405 5204
1406 6367
1407 3062
1410 6332
1411 5210
1412 6367
1413 2057
1414 3070
1415 1070
1416 7041
1417 1062
1420 7650
1421 5241
1422 1070
1423 7040
1424 1062
1425 7650
1426 5241
1427 1062
1430 7040
1431 1070
1432 7650
1433 5210
1434 1057
1435 7041
1436 1107
1437 7640
1440 5210
1441 1050
1442 3106
1443 4517
1444 2106
1445 5243
1446 4511

*. 17/+1
/TEST FOR NUMBER OF CHANNELS ON MULTIPLEXER
CHTST, CLA
DCA MUXNUM /CHANNEL ZERO
ACMX
ADCV /FIRST CONVERT
SKAD
JMP , -1
ICMX RAUC ADCV /SECOND CONVERT FIRST READ
DCA CTEMA
CHTST2, SKAD
JMP , -1
ICMX RAUC ADCV /3RD CONVERT 2ND READ
ISE MUXNUM
DCA BAD
TAU BAD
CIA
TAU CTEMA /FIRST READ
SNA CLA
JMP CHTST1
TAU BAD
CMA
TAU CTEMA /FIRST READ
SNA CLA
JMP CHTST1
TAU CTEMA
CMA
TAU BAD
SNA CLA /
JMP CHTST2 /TEST THE REST
TAU MUXNUM /NOT EQUAL +/- 1/2 LSB
CIA /CHANNEL
TAU K0040 /EQUAL TO MAXIMUM
SEA CLA
JMP CHTST2 /NO--NOT 40
CHTST1, TAU K0020
DCA STALL
JMS I XTEXTB /SAME VALUE IN X ICMX
ISE STALL
JMP , -2
SCOPE

```

```

/TEST CONTINGENCY REGISTER WAIT 36 MILL SEC
1447 7240 CLA CMA
1450 6334 XRCL /CLEAR CONTINGENCY CHANNEL
1451 7200 CLA
1452 2106 ISZ STALL /WAIT 36 MSEC FOR
1453 5252 JMP ,-1 /EXTERNAL STIMULUS
1454 2106 ISZ STALL
1455 5254 JMP ,-1
1456 6331 XRIN /READ CHANNELS
1457 7440 SZA /TEST FOR ZERO
1460 4512 HALT /EXTERNAL STIMULUS WAS APPLIED
1461 7240 CLA CMA
1462 6334 XRCL /CLEAR CHANNELS
1463 4511 SCOPE
1464 6354 CLRK

/LET S1, S2, OR S3 TRIGGER EXTERNAL INTERRUPT
/VALUE OF Y REGISTER CONSTANTLY INCREASING
/VALUE OF Y REGISTER IN AC IF INTERRUPT OCCUR

1465 1241 TAD K1004 /EXTERNAL INTERRUPT ENABLE
1466 6346 ZTEN OTEN /SELECT ENABLES
1467 1305 TAD VAL1 /INTERRUPT RETURN
1470 5202 DCA 2
1471 3106 DCA STALL /EQUAL (Y REGISTER)
1472 6201 ION /INTERRUPT ON
1473 6313 DYC DYI /LOAD Y REGISTER
1474 7200 NOP
1475 7200 NOP /TIME FOR INTERRUPT
1476 7201 IAC /INCREMENT (Y VALUE)
1477 2106 ISZ STALL /DONE ALL VALUES
1480 5273 JMP ,-5 /NO
1481 5303 JMP ,+2 /YES -- TURN INTERRUPT OFF
1482 4512 HALT /INTERRUPT OCCURED, AC=(Y-REGISTER)
1483 6202 IOP
1484 7410 SKP
1485 1502 VAL1, ,-3
1486 6342 ZTEN
1487 7240 CLA CMA
1490 6334 XRCL /CLEAR CONTINGENCY CHANNEL
1491 4511 SCOPE

/TEST THAT RC CLOCK DOES NOT INHIBIT ADCV
/ECG SEP 11,1968

1512 6354 CLRK
1513 6351 CLER
1514 1034 TAD K2002 /CLER CV AND CM
1515 6346 ZTEN OTEN /RC CLOCK INITIATE CONVERT
1516 6341 SKKK
1517 5316 JMP ,-1 /WAIT FOR FLAG
1520 6364 ADCV /NOW DO A ADCV, GENERATE ERROR
1521 6354 CLRK /CLEAR CLOCK FLAG
1522 6322 SKER /SKIP ON ERROR
1523 4512 HALT /ERROR NOT UP
1524 6342 ZTEN
1525 6332 SKAD

```

1226	5325	JMP , -1	/WAIT FOR FLAG
1227	6354	CLRK	
1230	6362	HAUC	/CLEAR A/D FLAG
1231	4511	SCOPE	
1232	6351	CLER	
1233	6342	ETEN	
/CRLF AT END OF PASS IF ERROR			
1234	7200	CLA	
1235	1513	TAU I XTYPE	/TEST FLAG
1236	7440	SZA	
1237	4516	JMS I XCRLF	/CRLF TO TTY
1240	3513	UCA I XTYPE	/CLEAR FLAG
1241	7504	LAS	
1242	7006	RTL	
1243	7710	SPA CLA	
1244	5521	JMP I XSWTCH	/TO DISPLAY TEST
1245	5746	JMP I , +1	/LOOP ON INSTRUCTION TEST
1246	0245	LOUPA	


```

1600 1600
1600 7402
1601 6351
1602 1060
1603 3104
1604 1044
1605 3105
1606 4515
1607 0057
1610 1635
1611 1635
1612 4454
1613 4543
1614 0411
1615 0107
1616 1617
1617 2324
1620 1103
1621 4022
1622 2516
1623 1611
1624 1607
1625 4543
1626 2301
1627 1505
1630 4026
1631 0114
1632 2505
1633 4011
1634 1640
1635 3030
1636 4011
1637 0315
1640 3000
1641 5600

*. 17/+1
TEXTB, XX
CLEAR
TAU K0003
UCA CHXL
TAU K0015
UCA CHYL
JMS I XSIXTY
MUNUM
TEXTB1
TEXTB1
JMS I MSSG /DISPLAY OUTPUT
4543 /DIAGNOSTIC RUNNING
0411
0107
1617
2324
1103
4022
2516
1611
1607
4543
2301 /SA
1505 /ME
4026 /SPACE V
0114 /AL
2505 /UE
4011 /SPACE I
1640 /N SPACE
TEXTB1, 3030 /XX
4011 /IC
0315 /MX
3000
JMP I TEXTB

```

1542	7402	SIXTY,	HLI	
1543	7000		NOP	
1544	7000		NOP	/STORE INIT NEXT TIME
1545	7200		CLA	
1546	1542		TAU I , -4	/ADDRESS OF OPERAND
1547	3251		UCA , +2	
1550	5652		JMP I , +2	
1551	0000		0	/ADDRESS OPERAND
1552	1654		SIXTY+12	/CHANGING REFERENCE (P)
1553	5245		JMP SIXTY+3	
1554	1651		TAU I SIXTY+7	/AC (OPERAND)
1555	0327		AND K0007	
1556	3322		UCA MASKA	/000X
1557	1651		TAU I SIXTY+7	/AC (OPERAND)
1560	0330		AND K0070	
1561	3323		UCA MASKB	/00X0
1562	1651		TAU I SIXTY+7	/AC (OPERAND)
1563	0331		AND K0700	
1564	3324		UCA MASKC	/0X00
1565	1651		TAU I SIXTY+7	/AC (OPERAND)
1566	0332		AND K7000	
1567	3325		UCA MASKD	/X000
1570	1324		TAU MASKC	/0X00
1571	7112		RTH CLL	
1572	7010		RAR	/0X00 RS3 00X0
1573	1325		TAU MASKD	/X0X0
1574	7012		RTH	
1575	7010		RAR	
1576	1326		TAU MASKD+1	/X0X0 RS3 0X0X
1577	3324		UCA MASKC	/TEMP STORAGE
1700	2242		ISZ SIXTY	/INCREMENT FOR STORAGE
1701	4252		JMS SIXTY+10	/FIND STORAGE ADDRESS
1702	1324		TAU MASKC	/6X6X
1703	3651		UCA I SIXTY+7	/STORE OPERAND AS SPECIFIED
1704	1323		TAU MASKB	/00X0
1705	7004		RAL	
1706	7006		RTL	/00X0 SL3 0X00
1707	1322		TAU MASKA	/0X00+000X=0X0X
1710	1326		TAU MASKD+1	/0X0X+6060=6X6X
1711	3325		UCA MASKD	/TEMP STORAGE ADDRESS
1712	2242		ISZ SIXTY	/INCREMENT FOR STORAGE
1713	4252		JMS SIXTY+10	/FIND STORAGE ADDRESS
1714	1325		TAU MASKD	/6X6X
1715	3651		UCA I SIXTY+7	/STORE OPERAND AS SECIFIED
1716	1333		TAU SIX12	/HOUSE KEEPING
1717	3252		UCA SIXTY+10	
1720	2242		ISZ SIXTY	/INCREMENT FOR RETURN
1721	5642		JMP I SIXTY	/RETURN

1/22	0000	MASKA,	0
1/23	0000	MASKB,	0
1/24	0000	MASKC,	0
1/25	0000	MASKD,	0
1/26	6060		6060
1/27	0007	K0007,	0007
1/30	0070	K0070,	0070
1/31	0700	K0700,	0700
1/32	7000	K7000,	7000
1/33	1654	SIX12,	SIXTY+12

```

2000 2000
2000 7604
2001 7104
2002 7530
2003 7410
2004 5237
2005 7040
2006 7530
2007 7410
2010 5243
2011 7060
2012 7730
2013 5247
2014 5271

2015 0000
2016 7604
2017 0131
2020 7106
2021 7006
2022 7006
2023 1132
2024 3225
2025 0000
2026 7604
2027 7106
2030 7710
2031 5200
2032 7604
2033 0107
2034 7640
2035 5510
2036 5615

2037 4326
2040 4523
2041 4215
2042 5237

2043 4522
2044 4524
2045 4215
2046 5243

2047 7300
2050 6303
2051 6313
2052 1142
2053 3143
2054 1140
2055 6307
2056 2143
2057 5254

*. 17/+1
SWTCH, LAS /DISCOVER WHAT TEST OPERATOR WANTS
      CLL RAL /MOVE SR0,1 INTO LINK AC FOR TEST
      SZL SPA
      SKP
      JMP LDIAG /HE WANTS LEFT DIAGONAL
      CMA
      SZL SPA
      SKP
      JMP RDIAG /HE WANTS RIGHT DIAGONAL
      CMA CML
      SZL SPA CLA
      JMP ANGLE /HE WANTS THE ANGLE
      JMP XBOX

KEYS, 0
      LAS
      AND K0600 /SAVE INTENSITY BITS
      CLL RTL /MOVE TO AC 10, 11
      RTL
      RTL /
      TAD K6324 /ADD BASIC IOT TO BITS
      DCA EXECUT /STORE
      EXECUT, 0 /EXECUTE
      LAS /GET SWITCHES
      CLL RTL
      SPA CLA /SHALL WE REPEAT
      JMP SWTCH /NO
      LAS
      AND K0040
      SZA CLA
      JMP I XINT3
      JMP I KEYS /YES

LDIAG, STUL /LEFT DIAGONAL SUBROUTINE
      DLUL /SET TOO UPPER LEFT
      JMS KEYS /DRAW LEFT DIAGONAL LINE
      JMP LDIAG /CHECK THE KEYS
      /REDRAW

RDIAG, STUR /SET TOO UPPER RIGHT
      DRUL /DRAW RIGHT DIAGONAL LINE
      JMS KEYS /CHECK THE KEYS
      JMP RDIAG /REDRAW

ANGLE, CLA CLL
      DXC DXL /CLEAR X
      DYC DYL /CLEAR Y
      TAD M200
      DCA COUNT /SET COUNTER
      TAD P2
      DXC DXL DIS /DRAW
      ISZ COUNT /DONE YET
      JMP ,-3 /NO

```

2060	7300	CLA CLL	
2061	1142	TAD M200	/SET UP COUNTER
2062	3143	DCA COUNT	
2063	1140	TAD P2	
2064	6317	DYC DYL DIS	/DRAW
2065	2143	ISZ COUNT	
2066	5263	JMP , -3	/DONE YET
2067	4215	JMS KEYS	/EXIT
2070	5247	JMP ANGLE	/REDRAW

2071	4316	XBOX, STLL
2072	4525	DVLT
2073	1133	TAD K377
2074	6303	DXC DXL
2075	3134	DCA XPOSR
2076	4526	DVLB
2077	4336	SJLR
2100	4525	DVLT
2101	6313	DYC DYL
2102	3135	DCA YPOSR
2103	4530	DHLL
2104	4326	STUL
2105	4523	DLUL
2106	4522	STUR
2107	4524	DRUL
2110	4326	STUL
2111	4527	DHLR
2112	4336	STLR
2113	4530	DHLL

2114	4215	JMS KEYS
2115	5271	JMP XBOX

2116	0000	SILLS,	0	CLA CLL	
2117	7300			UXC DXL	
2120	6303			UCA XPOSR	
2121	3134			IAD K400	
2122	1136			UYC DYL	/SET TO 0400
2123	6313			UCA YPOSR	
2124	3135			JMP I STLLS	
2125	5716				
2126	0000	STULS,	0	CLA CLL	
2127	7300			UXC	
2130	6301			UCA XPOSR	
2131	3134			IAD K377	
2132	1133			UYC DYL	
2133	6313			UCA YPOSR	
2134	3135			JMP I STULS	
2135	5726				
2136	0000	STLRS,	0	CLA CLL	
2137	7300			IAD K400	
2140	1136			UYC DYL	
2141	6313			UCA YPOSR	
2142	3135			IAD K777	
2143	1137			UXC DXL	
2144	6303			UCA XPOSR	
2145	3134			JMP I STLRS	
2146	5736				
2147	0000	STURS,	0	CLA CLL	
2150	7300			IAD K777	
2151	1137			UXC DXL	
2152	6303			UCA XPOSR	
2153	3134			IAD K377	
2154	1133			UYC DYL	
2155	6313			UCA YPOSR	
2156	3135			JMP I STURS	
2157	5747				
2160	0000	DVLTS,	0	IAD M400	
2161	1141			UCA COUNT	
2162	3143			IAD YPOSR	
2163	1135				

```

2164 1140      TAD P2
2165 6317      OYC DYI DIS
2166 2143      ISZ COUNT
2167 5364      JMP , -3
2170 7300      CLA CLL
2171 5760      JMP I DVLTS
          4316      STILL=JMS SILLS
          4326      STUL=JMS SIULS
          4336      SILR=JMS SILRS
          2200      *, 17/+1

```

```

2200 0000      DVLBS, 0
2201 1141      TAD M400
2202 3143      DCA COUNT
2203 1135      TAD YPOSR
2204 1052      TAD K7776
2205 6317      OYC DYI DIS
2206 2143      ISZ COUNT
2207 5204      JMP , -3
2210 7300      CLA CLL
2211 5600      JMP I DVLBS

```

/MINUS 2

```

2212 0000      DHLRS, 0
2213 1141      TAD M400
2214 3143      DCA COUNT
2215 1134      TAD XPOSR
2216 1140      TAD P2
2217 6307      OXC DXL DIS
2220 2143      ISZ COUNT
2221 5216      JMP , -3
2222 7300      CLA CLL
2223 5612      JMP I DHLRS

```

2224	0000	DHLLS,	0
2225	7300		CLA CLL
2226	1141		IAD M400
2227	3143		UCA COUNT
2230	1134		TAD XPOSR
2231	1052		TAD K7776
2232	6307		UXC DXL DIS
2233	2143		ISZ COUNT
2234	5231		JMP , -3
2235	7300		CLA CLL
2236	5624		JMP I DHLLS
2237	0000	DLCLS,	0
2240	7300		CLA CLL
2241	1141		TAD M400
2242	3143		UCA COUNT
2243	1134	MYPX,	TAD XPOSR
2244	1140		IAD P2
2245	6303		UXC DXL
2246	3134		UCA XPOSR
2247	1135		TAD YPOSR
2250	1052		TAD K7776
2251	6317		UYC DYL DIS
2252	3135		UCA YPOSR
2253	2143		ISZ COUNT
2254	5243		JMP MYPX
2255	5637		JMP I DLCLS
2256	0000	DRCLS,	0
2257	7300		CLA CLL
2260	1141		TAD M400
2261	3143		UCA COUNT
2262	4522		STUR
2263	1134	MYMX,	TAD XPOSR
2264	1052		TAD K7776
2265	6303		UXC DXL
2266	3134		UCA XPOSR
2267	1135		TAD YPOSR
2270	1052		TAD K7776
2271	6317		UYC DYL DIS
2272	3135		UCA YPOSR
2273	2143		ISZ COUNT
2274	5263		JMP MYMX
2275	5656		JMP I DRCLS

	2400		*, 177+1
2400	5205		JMP INIT3
	0010		PNTN=10
2401	0000		X, 0
2402	0000		Y, 0
2403	0000		RCNT, 0
2404	2514		ADBUFR, ADBUF=1
			pAUSE

```

/TAPE 3
/THREE STATE SCOPE DISPLAY
2405 7326 INIT3, CLA STL RTL /POSITIVE TWO
2406 6346      ZTEN OTEN
2407 6354      CLRK
2410 1047 INISAM, TAD KM1000
2411 3143      DCA COUNT
2412 1204      TAD ADBUFR
2413 3010      DCA PNTR
2414 7604 SAMLOP, LAS
2415 6375      ACMX ADCV /LOAD MUX WITH SWITCHES, CONVERT
2416 6332      SKAD
2417 5216      JMP , -1 /WAIT FOR FLAG
2420 6362      RAUC /READ BUFFER
2421 3410      DCA I PNTR /STORE SKP
/FOR SCOPING WITHOUT INDEXING MUX REPLAC PLACE A WITH A NOP
2422 5235      JMP SAM1
2423 7604      LAS /READ SWITCHES, SUBTRACT ONE
2424 1053      TAD K7777 /LOAD MUX, INCREMENT MVX
2425 6371      ACMX
2426 6361      ICMX /CONVERT
2427 6364      ADCV
2430 6332      SKAD
2431 5230      JMP , -1 /WAIT FOR FLAG
2432 6362      RAUC /READ BUFFER
2433 3410      DCA I PNTR /STORE
2434 2143      ISZ COUNT
2435 2143 SAM1, ISZ COUNT
2436 5214      JMP SAMLOP /MORE SAMPLES
2437 1050      TAD KM0020
2440 3203      DCA RCNT

2441 1204 INIDIS, TAD ADBUFR /BUFFER
2442 3010      DCA PNTR
2443 1024      TAD KM0777

2444 3143      DCA COUNT
2445 3201      DCA X /X REFERENCE SETUP
2446 1410      TAD I PNTR
2447 7041      CMA IAC
2450 3202      DCA Y /Y REFERENCE SETUP

```

2451	4276	JMS STATE	/VALUE OF CONVERTED WORD
2452	1410	DISLOP, TAD I PNTR	/NEXT WORD
2453	1202	TAD Y	/REFERENCE IT
2454	7006	RTL	
2455	7006	RTL	
2456	7006	RTL	
2457	0045	AND K7700	
2460	0313	OYC DYI	/LOAD Y
2461	7200	CLA	
2462	1201	TAD X	
2463	6307	OXC OXL DIS	/LOAD X
2464	7001	IAC	
2465	3201	OCA X	/RESTORE X INCREMENTED
2466	2143	ISZ COUNT	/DONE
2467	5252	JMP DISLOP	/NO
2470	6341	SKRK	/SKIP ON RC CLOCK
2471	5241	JMP INIDIS	/RC CLOCK IS REFRESH RATE
2472	6354	CLRK	/CLEAR CLOCK
2473	2203	ISZ RCNT	/BASIC TIME COUNT
2474	5241	JMP INIDIS	/REFRESH DISPLAY
2475	5210	JMP INISAM	/FILL BUFFER WITH NEW
2476	7402	STATE, XX	
2477	6351	CLER	
2500	1060	TAD K0003	
2501	3104	OCA CHXL	
2502	1042	TAD K0340	
2503	3105	OCA CHYL	
2504	4515	JMS I XSIXTY	
2505	2515	A0BUF	
2506	2511	,+3	
2507	2512	,+3	
2510	4454	JMS I MSSG	
2511	6060	0000	
2512	6060	0000	
2513	0000	0	
2514	5676	JMP I STATE	
2515	0000	A0BUF, 0	
	5515	*A0BUF+1000	

3606

*. 17/+1

/FILE NAME: DSCPAK

/DISPLAY STRIPPED ASCII CHAR IN AC: DSC

/SAMPLE CALLING SEQUENCE -

/ LDM /GET HALFWORD

/ DSC /DISPLAY IF NON ZERO

/ JMP ,-2 /DISPLAY THE NEXT HALFWORD

4455

DSC=JMS I CHDSCI

/TEMPORARY STORAGE:

00/6

CHCNT3=TEMP02

00/7

CHCNT2=TEMP03

0100

CHCNT1=TEMP04

0101

CHROT=TEMP05

02/5

CHAC=TEMP01

0102

CHPNT=TEMP06

02/5

CHYS=TEMP01

/DISPLAY TABLE EXCEPTIONS

7777

SPACE=-1

0205

ENDIS=5

0201

BKSPC=1

0202

RESET=2

0203

LFEE0=3

0204

CRETN=4

0200

IGNOR=0

```

600 0000 CHUSC, 0
601 7104 CLL RAL
602 1056 TAU ADSTBL
603 3102 UCA CHPNT
604 1502 TAU I CHPNT
605 2102 ISZ CHPNT
606 7440 SZL
607 5223 JMP CHSET1 /NOT SPECIAL CHARACTER
610 1502 TAU I CHPNT
611 1360 TAU CHJMP
612 3213 UCA ,+1
613 5600 JMP I CHUSC

```

```

614 3676 /DISPATCH TABLE FOR ABOVE JMP
615 3671 CHJMPL, CHSPA /JUMP TABLE: SPACE
616 3700 CHEND+1 /IGNORE
617 3715 CHBKS /BACKSPACE
620 3751 CHREST /NEXT 4 CHARACTERS RESET X, Y, DELTA, INTENSITY
621 3747 CHLF /DO A LINE FEED
622 3670 CHCR /DO A CARRIAGE RETURN
CHEND /END OF TEXT

```

```

623 3101 CHSET1, UCA CHROT
624 1104 TAU CHXL
625 6303 DXC DXL
626 7344 MTW /NUMBER OF WORDS IS TWO
627 3076 UCA CHCNT3
630 7344 CHSET2, MTW /NUMBER OF LINES IS TWO
631 3077 UCA CHCNT2
632 1061 CHSET3, TAU KM0006 /NUMBER OF POINTS IN A LINE IS SIX
633 3100 UCA CHCNT1
634 1105 TAU CHYL /REPOSITION Y AT BOTTOM OF CHARACTER
635 3075 UCA CHYS /RECORD PRESENT Y POSITION
636 1101 CHDIS, TAU CHROT
637 7104 CLL RAL /NO, GET FIRST BIT INTO LINK
640 3101 UCA CHROT /STORE CODE WORD FOR NEXT BIT
641 1075 TAU CHYS /INTENSIFICATION POSITION
642 6313 OYC DYL
643 7430 SZL /DISPLAYS IF LINK=1
644 6304 DIS
645 1361 TAU CHSI2 /GO NEXT POSSIBLE DOT-
646 3075 UCA CHYS /KEEP RECORD OF PRESENT POSITION
647 2100 ISZ CHCNT1 /DO ALL THIS SIX TIMES
650 5236 JMP CHDIS
651 1104 TAU CHXL
652 1361 TAU CHSI2
653 6303 DXC DXL
654 3104 UCA CHXL
655 2077 ISZ CHCNT2 /HAS SECOND HALF BEEN DONE?
656 5232 JMP CHSET3 /NO, DO IT
657 1502 TAU I CHPNT /GET SECOND WORD
660 3101 UCA CHROT
661 2076 ISZ CHCNT3
662 5230 JMP CHSET2

```

9/26/68 17:18,37

PAGE 35-1

3563 1361
3564 7104
3565 1104
3566 3104
3567 5600

TAD CHSIZ /BOTH DONE
CLL RAL
TAD CHXL
UCA CHXL /SPACE 2 INCREMENTS
JMP I CHDSC

35/0 2200
35/1 6303
35/2 1364
35/3 6313
35/4 7200
35/5 5500

CHEND, ISZ CHDSC
DXC DXL
TAD K0401
DYC DYL
CLA
JMP I CHDSC

35/6 4305
35/7 5302

/SPACE: MOVE X RIGHT 6 INCREMENTS
CHSPA, JMS GET6 /GET X FOR WHOLE CHARACTER
JMP CHRKS+2

3/00 4305
3/01 7041
3/02 1104
3/03 3104
3/04 5500

/BACKSPACE: MOVE X LEFT ONE CHARACTER
CHBKS, JMS GET6
CMA IAC
TAD CHXL /MOVE X POINTER LEFT
JCA CHXL /ONE CHARACTER
JMP I CHDSC

3/05 0000
3/06 1361
3/07 7106
3/10 3075
3/11 1361
3/12 7104
3/13 1075
3/14 5705

/GETS SIX TIMES CHAR SIZE
GET6, 0
TAD CHSIZ
CLL RTL /4 TIMES SIZE
JCA CHFAC
TAD CHSIZ
CLL RAL /PLUS 2 TIMES SIZE
TAD CHFAC /IS SIX TIMES SIZE
JMP I GET6

3/15 4454
3/16 1362
3/17 0363
3/20 7106
3/21 7006
3/22 7004
3/23 3104

/SELECT NEW X POSITION
CHREST, LDH
TAD K0001
AND K0017
CLL RTL
RTL
RAL
JCA CHXL

3/24 4454
3/25 0363
3/26 7106
3/27 7006
3/30 7004
3/31 7041
3/32 1364
3/33 3105

/SELECT NEW Y POSITION
CHYPOS, LDH
AND K0017
CLL RTL
RTL
RAL
CMA IAC
TAD K0401
JCA CHYL

/CHARACTER SIZE SELECTION

3/34	4454	CHCSZ,	LDM	
3/35	0363		AND K0017	/ELIMINATE ALL BUT BITS 8-11
3/36	7104		CLL RAL	
3/37	3361		UCA CHSIZ	/STORE DESIRED CHARACTER SIZE

/INTENSIFICATION SELECTION

3/40	4454	CHINT,	LDM	
3/41	0060		AND K0003	/ELIMINATE ALL BUT BITS 10 AND 11
3/42	1365		IAD KDSB	/SET INTENSITY
3/43	3344		UCA ,+1	
3/44	6324		USB	/WILL BE DSB X
3/45	5600		JMP I CHDSC	
3/46	5600		JMP I CHDSC	

/CARRIAGE RETURN: RETURN X TO LEFT MARGIN

3/47	3104	CHCR,	UCA CHXL	
3/50	5600		JMP I CHDSC	

/LINE FEED: MOVE Y DOWN 8 INCREMENTS

3/51	1361	CHLF,	IAD CHSIZ	/8 TIMES CHAR SIZE
3/52	7106		CLL RTL	
3/53	7004		RAL	
3/54	7041		CMA IAC	/SUBTRACT FROM Y POSITION
3/55	1105		IAD CHYL	
3/56	3105		UCA CHYL	/NEW Y POSITION
3/57	5600		JMP I CHDSC	

/CONSTANTS USED THIS PAGE ONLY

3/60	5615	CHJMP,	JMP I CHJMPL+1
------	------	--------	----------------

/VARIABLES FOR THIS PAGE

3/61	0004	CHSIZ,	4	/CHARACTER SIZE CONSTANT
------	------	--------	---	--------------------------

3/62	7777	KM001,	-1
3/63	0017	K0017,	17
3/64	0401	K0401,	401
3/65	6324	KDSB,	DSB

	USCTBL. 0	/@ SPECIAL, MEANS END OF TEXT
3/66 0000	ENDIS	
3/67 0005		
3/70 7711	//11	/A
3/71 1177	1177	
3/72 7745	//45	/B
3/73 4532	4532	
3/74 3641	3641	/C
3/75 4122	4122	
3/76 7741	//41	/D
3/77 4136	4136	
4000 7751	//51	/E
4001 5141	5141	
4002 7711	//11	/F
4003 1101	1101	
4004 3641	3641	/G
4005 5132	5132	
4006 7710	//10	/H
4007 1077	1077	
4010 0077	0077	/I
4011 0000	0000	
4012 2040	2040	/J
4013 4037	4037	
4014 7714	//14	/K
4015 2241	2241	
4016 7740	//40	/L
4017 4040	4040	
4020 7706	//06	/M
4021 0677	0677	
4022 7704	//04	/N
4023 1077	1077	
4024 7741	//41	/O
4025 4177	4177	
4026 7711	//11	/P
4027 1106	1106	
4030 1621	1621	/Q
4031 3156	3156	

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13	711	7711	/R
13	3146	3146	
134	2245	2245	/S
135	4531	4531	
13	0101	0101	/T
137	7701	7701	
14	7740	7740	/U
14	4077	4077	
142	3740	3740	/V
142	2017	2017	
14	7730	7730	/W
145	4077	4077	
14	6314	6314	/X
14	1463	1463	
050	0770	0770	/Y
05	7007	7007	
052	6151	6151	/Z
053	4543	4543	
05	0617	0617	/L DISPLAYED AS #
055	1706	1706	
055	1057	1057	/\ DISPLAYED AS DOWN ARROW
0	1000	1000	
060	2313	2313	/J DISPLAYED AS %
061	6462	6462	
0 2	0475	0475	/?
063	0400	0400	
/SPECIAL, DOES BACKSPACE			
0 4	0000	0	
065	0001	BACKSPACE	
0 5	0000	0	/SPACE IS A SPECIAL CHARACTER
0 7	7777	SPACE	
070	0057	0057	/!
071	0000	0000	
0 2	0000	0	/" SPECIAL, MARKS PROG INPUT TO TEXT BUFFER, NOT DISPL
073	0000	IGNOR	
0 4	0000	0	/# SPECIAL, DOES LINE FEED
0 5	0003	LFEEED	

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0000	0	/S SPECIAL, DISPLAY RESET FOLLOWS
002	RESET	
0000	0	/X SPECIAL, DOES CARRIAGE RETURN
01 0004	CRETN	
0017	0017	/8
0010	0010	
0000	0	/I SPECIAL, MARKS KBD INPUT TO TEXT BUFFER, NOT DISPLA
0000	IGNOR	
0041	0041	/C DISPLAYED AS [
0100	0100	
0041	0041	/) DISPLAYED AS]
0100	0100	
2214	2214	/* DISPLAYED AS LITTLE X
1422	1422	
0416	0416	/+
0400	0400	
0050	0050	/,
5000	5000	
0404	0404	/-
0400	0400	
0040	0040	/.
0000	0000	
010	0010	//
0402	0402	
3641	3641	/0
4136	4136	
0042	0042	/1
0740	0740	
6251	6251	/2
5146	5146	
2241	2241	/3
4532	4532	
1412	1412	/4
7710	7710	
2745	2745	/5
4531	4531	

4142	3645	3645	/6
4143	4530	4530	
4144	6111	6111	/7
4145	0503	0503	
4146	3245	3245	/8
4147	4532	4532	
4150	0651	0651	/9
4151	5136	5136	
4152	0033	0033	/:
4153	0000	0000	
4154	0053	0053	/;
4155	3000	3000	
4156	1422	1422	/<
4157	4100	4100	
4160	0012	0012	/=
4161	1212	1212	
4162	4122	4122	/>
4163	1400	1400	
4164	0251	0251	/?
4165	0502	0502	
4200			

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/LOAD HALF WORD AN DISPLAY
MSG, XX

4200	7402	CLA CMA
4201	7240	TAU MSG
4202	1200	DCA 10
4203	3010	TAU I 10
4204	1410	DCA MSRHT
4205	3103	TAU MSRHT
4206	1103	RTR
4207	7012	RTR
4210	7012	RTR
4211	7012	AND K0077
4212	0046	USC
4213	4455	SKP CLA
4214	7610	JMP I 10
4215	5410	TAU MSRHT
4216	1103	AND K0077
4217	0046	USC
4220	4455	JMP MSG+4
4221	5204	JMP I 10
4222	5410	

/SCOPE LOOP CHECK
 /SCOPE WITH SWITCH BIT 10=1
 SCOPEA, XX

4223	7402	LAS
4224	7604	AND K0002
4225	0033	SEA CLA
4226	7640	JMP I RETURN
4227	5533	IAU SCOPEA
4230	1223	JCA RETURN
4231	3233	JMP I SCOPEA
4232	5523	

4233	0200	RETURN, BEGIN
4234	5533	JMP I RETURN

/INHIBIT PRINT OUT WHEN BIT 11=1

4235	7402	IPRINT, XX	
4236	3120	UCA AC	/SAVE AC
4237	7504	LAS	
4240	7010	RAR	
4241	7620	SNL CLA	
4242	5250	JMP ,+6	
4243	1230	IAU IPRINT	
4244	1052	IAU K/7/6	/MINUS 2
4245	3230	UCA IPRINT	
4246	1530	IAU I IPRINT	
4247	3230	UCA IPRINT	
4250	1120	IAU AC	/RESTORE AC
4251	5530	JMP I IPRINT	

/GENERATE CR,LF

4252	7402	CRLF, XX
4253	4013	JMS I XIYPE
4254	4043	4543
4255	0000	0000
4256	0032	KCC
4257	0042	ICF
4260	5552	JMP I CRLF

/PRINTOUT CONTENTS OF AC AND ERROR ADDRESS

4261	7402	ERAOD, XX	
4262	4514	JMS I XIPRIN	/TEST FOR INHIBIT PRINT
4263	6002	ICF	
4264	4515	JMS I XSIXTY	
4265	4261	ERAOD	
4266	4272	,+4	
4267	4273	,+4	
4270	4513	JMS I XTYPE	/TYPE ERROR ADDRESS
4271	4543	4543	
4272	6060	6060	
4273	6060	6060	
4274	4000	4000	
4275	4515	JMS I XSIXTY	
4276	0120	AC	
4277	4303	,+4	
4300	4304	,+4	
4301	4513	JMS I XTYPE	/TYPE AC
4302	4040	4040	
4303	6060	6060	
4304	6060	6060	
4305	0000	0000	
4306	6042	ICF	
4307	6032	KCC	
4310	7604	LAS	/TEST FOR HALT
4311	7012	RTR	/BIT 9=1, (0004)
4312	7010	RAR	
4313	7620	SNL CLA	
4314	5601	JMP I ERAOD	
4315	1120	IAU AC	
4316	7402	HLI	
4317	5601	JMP I ERAOD	

PAUSE

/TAPE 4
 /DIGITAL 8-18-U
 /MESSAGE TYPE-OUT
 /CALL WITH A JMS MESSAGE
 /WITH DATA FOLLOWING
 /RETURN FOLLOWING END OF MESSAGE
 /CODE(00)

4320	0000	MESSAGE,	0	
4321	7240		CLA CMA	/SET C(AC)=-1
4322	1320		IAD MESSAGE	/ADD LOCATION
4323	3010		JCA 10	/AUTO-INDEX REGISTER
4324	1410		IAD I 10	/FETCH FIRST WORD
4325	3336		JCA MSRGHT	/SAVE IT
4326	1336		IAD MSRGHT	
4327	7012		RIR	
4330	7012		RTR	/ROTATE 6 BITS RIGHT
4331	7012		RTR	
4332	4337		JMS TYPECH	/TYPE IT
4333	1336		IAD MSRGHT	/GET DATA AGAIN
4334	4337		JMS TYPECH	/TYPE RIGHT HALF
4335	5324		JMP MESSAGE+4	/CONTINUE
4336	0000	MSRGHT,	0	/TEMPORARY STORAGE
4337	0000	TYPECH,	0	/TYPE CHARACTER IN C(AC)6-11
4340	0370		AND MASK77	
4341	7450		SNA	/IS IT END OF MESSAGE?
4342	5410		JMP I 10	/YES: EXIT
4343	1371		IAD M40	/SUBTRACT 40
4344	7500		SMA	/<40?
4345	5350		JMP ,+3	/NO
4346	1372		IAD C340	/YES: ADD 300
4347	5363		JMP MTP	/TO CODES <40
4350	1373		IAD M3	/SUBTRACT 3
4351	7440		SZA	/IS IT ZERO?
4352	5355		JMP ,+3	/NO
4353	1374		IAD C212	/YES: CODE 43 IS
4354	5363		JMP MTP	/LINE-FEED (212)
4355	1375		IAD M2	/SUBTRACT 2
4356	7440		SZA	/IS IT ZERO?
4357	5362		JMP ,+3	/NO
4360	1376		IAD C215	/YES: CODE 45 IS
4361	5363		JMP MTP	/CARRIAGE-RETURN (215)
4362	1377		IAD C245	/ADD 200 TO OTHERS >40
4363	6046	MTP,	ILS	/TRANSMIT CHARACTER
4364	6041		ISF	/WAIT FOR FLAG
4365	5364		JMP , -1	/NOT SET YET
4366	7200		CLA	/SET: CLEAR C(AC)
4367	5737		JMP I TYPECH	/RETURN

4370	0077	/CONSTANTIS	
4371	7740	MASK77,	77
4372	0340	M40,	-40
4373	7775	C340,	340
4374	0212	M3,	-3
4375	7776	C212,	212
4376	0215	M2,	-2
4377	0245	C215,	215
		C245,	245

3

THERE ARE NO ERRORS

SYMBOL TABLE

AC	0120
ACMA	6371
ADHUF	2515
ADHUFK	2404
AUCV	6364
AUSIBL	0056
ANGLE	2047
BAU	0070
BEGLY	0200
BASPC	0001
CCONTA	0065
CCONTB	0066
CCONTC	0067
CHBAS	3700
CHCNT1	0100
CHCNT2	0077
CHCNT3	0076
CHCR	3747
CHCS	3734
CHDIS	3636
CHDSC	3600
CHDSCI	0055
CHEND	3670
CHFAC	0075
CHINT	3740
CHJMP	3750
CHJMPL	3614
CHLF	3751
CHPNT	0102
CHRES1	3715
CHROI	0101
CHSET1	3623
CHSET2	3630
CHSET3	3632
CHSLX	3761
CHSPA	3676
CHTST	1400
CHTST1	1441
CHTST2	1410
CHXL	0104
CHYL	0105
CHYPOS	3724
CHYS	0075
CLER	6351
CLRK	6354
CLXK	6352
CUN1	1340
CUN2	1353
CUN3	1331
CUUNT	0143
CRETN	0004
CRLF	4252
CIEMA	0062

SYMBOL TABLE

CIEMR	0063
CIEMC	0064
COO70	0037
C212	4374
C215	4376
C245	4377
C340	4372
UHLL	4530
UHLLA	0130
UHLLS	2224
UHLLR	4527
UHLLRA	0127
UHLLRS	2212
UIS	6324
UISLOP	2452
ULUL	4523
ULDLA	0123
ULDLS	2237
URDL	4524
URDLA	0124
URDLS	2256
USH	6324
USC	4455
USCIBL	3766
UVLB	4526
UVLBA	0126
UVLBS	2200
UVLI	4525
UVLIA	0125
UVLIS	2160
UAC	6341
JAL	6302
JYC	6311
JYL	6312
ENDIS	0005
EMADD	4261
EXECUI	2025
FI1	0026
GET6	3705
HALI	4512
ICMX	6361
IGNOR	0000
INIDIS	2441
INISAM	2410
INIT3	2405
INLL	0071
IPRINI	4235
KUSB	3765
KEYS	2015
KM0001	3762
KM0006	0061
KM0020	0050
KM0043	0022

SYMBOL TABLE

KMP777	0024
KM1000	0047
K0002	0033
K0003	0050
K0004	0036
K0007	1/27
K0015	0044
K0017	3/63
K0034	0036
K0043	0107
K0070	1/30
K0077	0046
K0103	0031
K0200	0040
K0340	0042
K0400	0023
K0401	3/64
K0600	0131
K0700	1/31
K1000	0032
K1002	0051
K1004	0041
K2002	0034
K3002	0035
K377	0133
K400	0136
K4002	0043
K0324	0132
K1000	1/32
K1400	0021
K1700	0045
K177	0137
K1776	0052
K1777	0053
LUH	4454
LUIAG	2037
LFEEJ	0003
LUOP	0072
LUOPA	0245
MASKA	1/22
MASKB	1/23
MASKC	1/24
MASKD	1/25
MASK77	4370
MESSAGE	4320
MSG	4200
MSRGHI	4336
MSRHT	0103
MSSG	0054
MIP	4363
MIW	7344
MUXNUM	0057
MYMX	2263

SYMBOL TABLE

MYPX	2243
M0377	0027
M2	4375
M200	0142
M3	4373
M40	4371
M400	0141
M/400	0020
UPR	7000
UIEN	6344
PNTK	0010
P2	0140
RADC	6352
RUNT	2403
RUX	1200
RUX1	1241
RUX2	1303
RUX3	1324
RUIAG	2043
RESET	0002
RETURN	4233
SAMLOP	2414
SAM1	2435
SCOPE	4511
SCOPEA	4223
SIX1Y	1642
SIX12	1733
SKAD	6332
SKER	6322
SKRK	6341
SKXK	6321
SL1	0025
SPACE	7777
SIAL	0106
SIATE	2476
SILL	4316
SILLS	2116
SILR	4336
SILKS	2136
SIUL	4326
SIULS	2126
SIUR	4522
SIURA	0122
SIURS	2147
SWTCH	2000
TEMP01	0075
TEMP02	0076
TEMP03	0077
TEMP04	0100
TEMP05	0101
TEMP06	0102
EXT0	1600
EXT01	1635

SYMBOL TABLE

TYPECH	4337
VAL1	1505
X	2401
XBOX	2071
XCRLE	0116
XCR	0074
XERRADU	0112
XIN13	0110
XIPRIN	0114
XPOSR	0134
XACL	6334
XRLN	6331
XSCOPE	0111
XSIXY	0115
XSWICH	0121
XLEXIB	0117
XTYPE	0113
XX	7402
Y	2402
YPOSR	0135
ZLEN	6342

SYMBOL TABLE

IGNOR	0000
BASPC	0001
RESET	0002
LFEEJ	0003
CHETN	0004
ENDIS	0005
PVTR	0010
M/400	0020
K/400	0021
KM0040	0022
K0400	0023
KM0777	0024
SL1	0025
FL1	0026
M0377	0027
K0034	0030
K0100	0031
K1000	0032
K0002	0033
K2002	0034
K3002	0035
K0004	0036
C0070	0037
K0200	0040
K1004	0041
K0340	0042
K4002	0043
K0015	0044
K7000	0045
K0077	0046
KM1000	0047
KM0020	0050
K1002	0051
K7776	0052
K7777	0053
MSSG	0054
CHDSCI	0055
AUSTBL	0056
MUXNUM	0057
K0003	0060
KM0000	0061
CIEMA	0062
CIEMB	0063
CIEMC	0064
CCONTA	0065
CCONTB	0066
CCONTC	0067
BAD	0070
INLL	0071
LUOP	0072
XER	0074
CHYS	0075
TEMP01	0075

SYMBOL TABLE

CMFAC	0075
CMCN13	0076
TEMP22	0076
CMCN12	0077
TEMP23	0077
CMCN11	0100
TEMP24	0100
TEMP25	0101
CMR01	0101
CMR01	0102
TEMP26	0102
MSR01	0103
CMXL	0104
CMYL	0105
SIALL	0106
K0040	0107
XIN13	0110
XSCOPE	0111
XERADU	0112
XIYPE	0113
XIPRIN	0114
XSIXY	0115
XCLRF	0116
XIEX1B	0117
AC	0120
XSWICH	0121
STORA	0122
ULOLA	0123
UROLA	0124
UVLIA	0125
UVLBA	0126
UHLKA	0127
UHLLA	0130
K0602	0131
K0324	0132
K377	0133
XPOS2	0134
YPOS2	0135
K400	0136
K777	0137
P2	0140
M400	0141
M200	0142
CUUN1	0143
BEGIV	0200
LUOPA	0245
RUX	1200
RUX1	1241
RUX2	1303
RUX3	1324
CUN3	1331
CUN1	1340
CUN2	1353

SYMBOL TABLE

CHTST	1400
CHTST2	1410
CHTST1	1441
VAL1	1505
TEXTB	1600
TEXTB1	1635
SIXTY	1642
MASKA	1722
MASKB	1723
MASKC	1724
MASKD	1725
K0007	1727
K0072	1730
K0702	1731
K7002	1732
SIX12	1733
SWTCH	2000
KEYS	2015
EXECUT	2025
CU1AG	2037
KU1AG	2043
ANGLE	2047
XBOX	2071
SILLS	2116
SIOLS	2126
SILRS	2136
SIORS	2147
UVLIS	2150
UVLBS	2200
UHLRS	2212
UHLLS	2224
ULDLS	2237
MTPX	2243
URDLS	2256
MYMX	2263
X	2401
Y	2402
RCN1	2403
AUBUFK	2404
INIT3	2405
INISAM	2410
SAMLOP	2414
SAM1	2435
INIDIS	2441
DISLOP	2452
SIATE	2476
AUBUF	2515
CHDSC	3600
CHJMPL	3614
CHSET1	3623
CHSET2	3630
CHSET3	3632
CHDIS	3636

SYMBOL TABLE

CHEND	3670
CHSPA	3676
CHBKS	3700
GET6	3705
CHRES1	3715
CHYPOS	3724
CHCS2	3734
CHINT	3742
CHCR	3747
CHLF	3751
CHJNP	3750
CHSIZ	3751
KM0001	3752
K0017	3753
K0401	3754
KUSB	3755
USCIBL	3756
MESG	4200
SCOPEA	4223
RETURN	4233
IPRINT	4235
CHLF	4252
ERADD	4251
SILL	4316
MESSAGE	4320
SIUL	4326
SILK	4336
MORCHI	4336
TYPECH	4337
MIP	4363
MASK77	4370
140	4371
U340	4372
M3	4373
U212	4374
M2	4375
U215	4376
U245	4377
LUH	4454
USC	4455
SCOPE	4511
HALT	4512
SIUR	4522
UL0L	4523
UK0L	4524
UVLT	4525
UVLB	4526
UHLR	4527
UHLL	4530
UXC	6301
UXL	6302
UIS	6304
UYC	6311

SYMBOL TABLE

DTL	6312
SXXX	6321
SXER	6322
JSB	6324
XKIN	6331
SXAD	6332
XKCL	6334
SXRR	6341
TIEN	6342
UIEN	6344
CLER	6351
CLXA	6352
CLRX	6354
ICMX	6361
RADU	6362
AUCV	6364
ACMX	6371
JPR	7000
MIW	7344
XA	7402
SPACE	7777

